

Privacy-Preserving Machine Learning on Web Browsing for Public Opinion

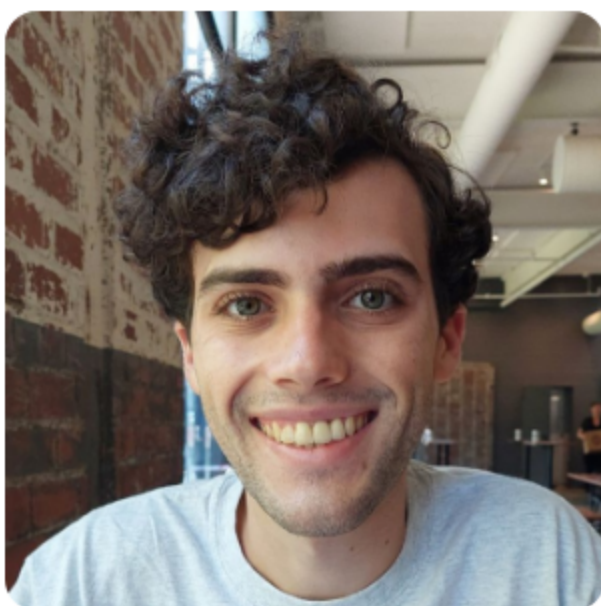
Sam Buxbaum

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CSCML 25

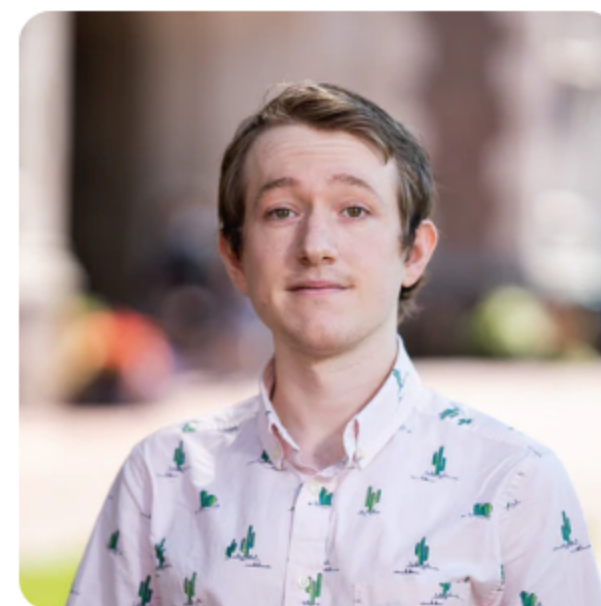
December 4, 2025



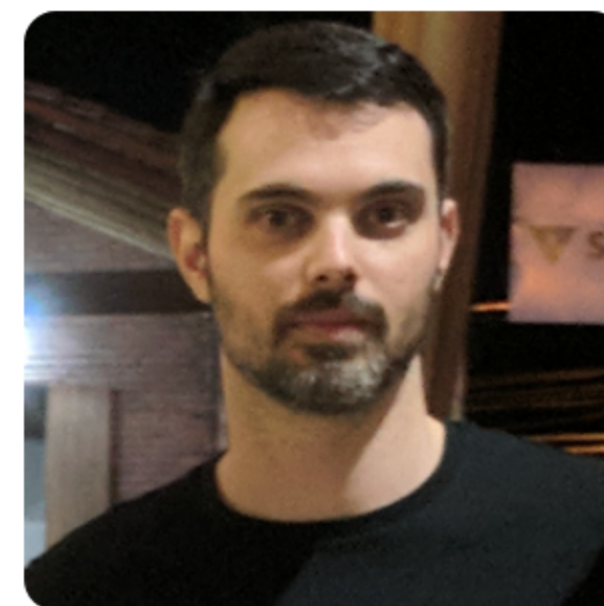
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Traditional Political Polling

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West Virginia 2024 Presidential Election Polls



Harris vs. Trump

Source	Date	Sample	Harris	Trump	Other
Research America	8/30/2024	400 LV $\pm$ 4.9%	34%	61%	5%

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Michigan 2024 Presidential Election Polls



☐ Instantly compare a poll to prior one by same pollster

Harris vs. Trump

Source	Date	Sample	Harris	Trump	Other
Average of 23 Polls†			48.6%	46.8%	-
FAU / Mainstreet	11/04/2024	713 LV	49%	47%	4%
Emerson College	11/04/2024	790 LV $\pm$ 3.4%	50%	48%	2%
Research Co.	11/04/2024	450 LV $\pm$ 4.6%	49%	47%	4%
InsiderAdvantage	11/03/2024	800 LV $\pm$ 3.7%	47%	47%	6%
Trafalgar Group	11/03/2024	1,079 LV $\pm$ 2.9%	47%	48%	5%
MIRS / Mich. News Source	11/03/2024	585 LV $\pm$ 4%	50%	48%	2%
NY Times / Siena College	11/03/2024	998 LV $\pm$ 3.7%	47%	47%	6%
Morning Consult	11/03/2024	1,108 LV $\pm$ 3%	49%	48%	3%
AtlasIntel	11/02/2024	1,198 LV $\pm$ 3%	48%	50%	2%
Redfield & Wilton	11/01/2024	1,731 LV $\pm$ 2.2%	47%	47%	6%
The Times (UK) / YouGov	11/01/2024	942 LV $\pm$ 3.9%	48%	45%	7%
EPIC-MRA	11/01/2024	600 LV $\pm$ 4%	48%	45%	7%
Marist Poll	11/01/2024	1,214 LV $\pm$ 3.5%	51%	48%	1%
AtlasIntel	10/31/2024	1,136 LV $\pm$ 3%	49%	49%	2%
Echelon Insights	10/31/2024	600 LV $\pm$ 4.4%	48%	48%	4%
MIRS / Mich. News Source	10/31/2024	1,117 LV $\pm$ 2.5%	47%	49%	4%
UMass Lowell	10/31/2024	600 LV $\pm$ 4.5%	49%	45%	6%
Washington Post	10/31/2024	1,003 LV $\pm$ 3.7%	47%	46%	7%
Fox News	10/30/2024	988 LV $\pm$ 3%	49%	49%	2%
CNN	10/30/2024	726 LV $\pm$ 4.7%	48%	43%	9%
Suffolk University	10/30/2024	500 LV $\pm$ 4.4%	47%	47%	6%

Web Browsing for Political Polling

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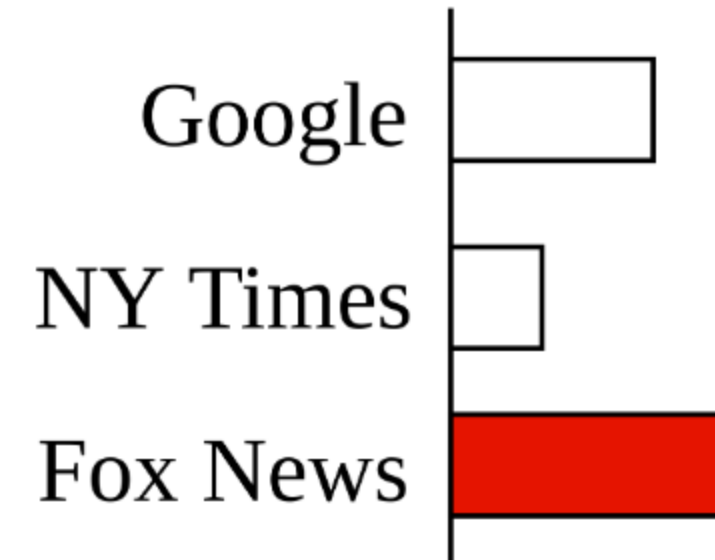
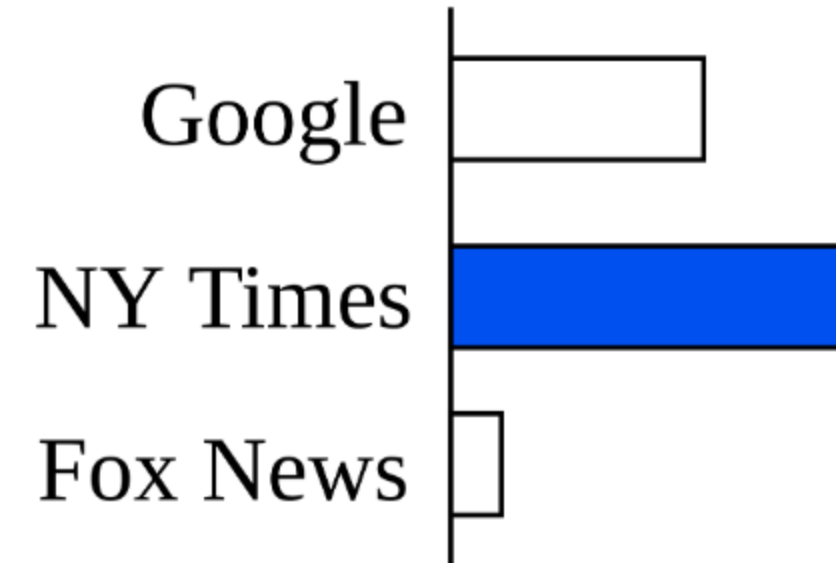
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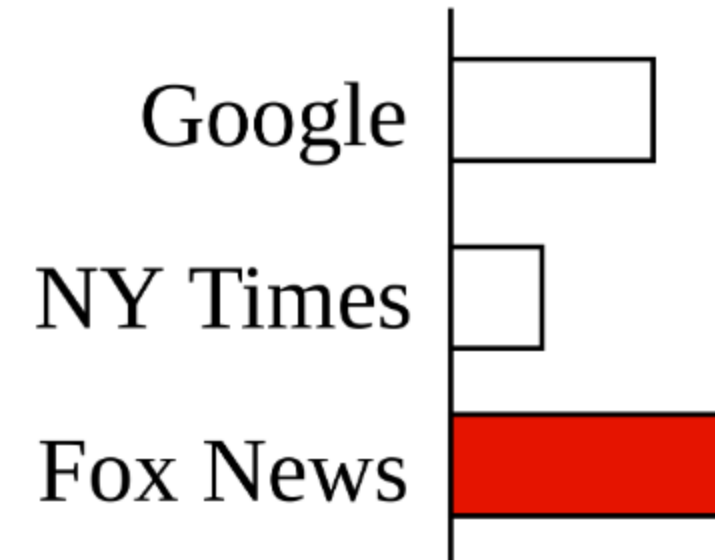
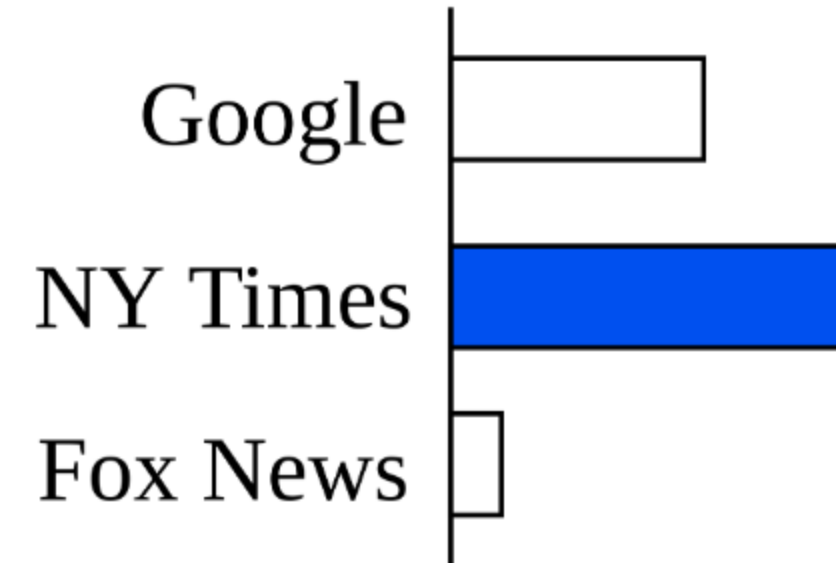
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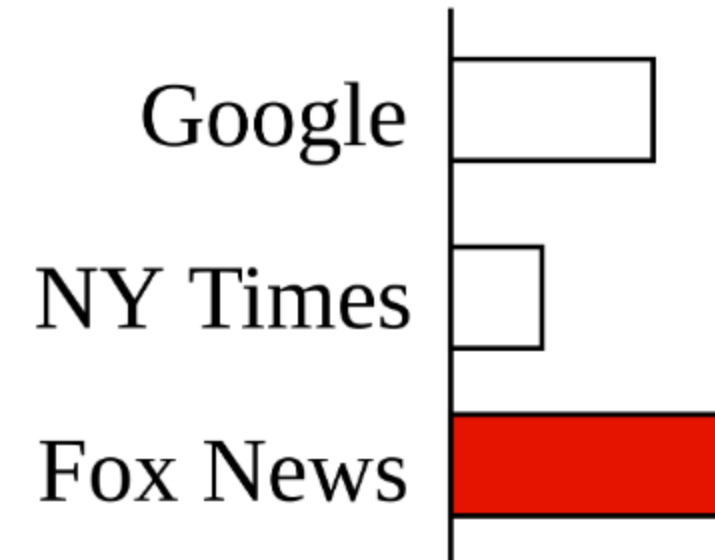
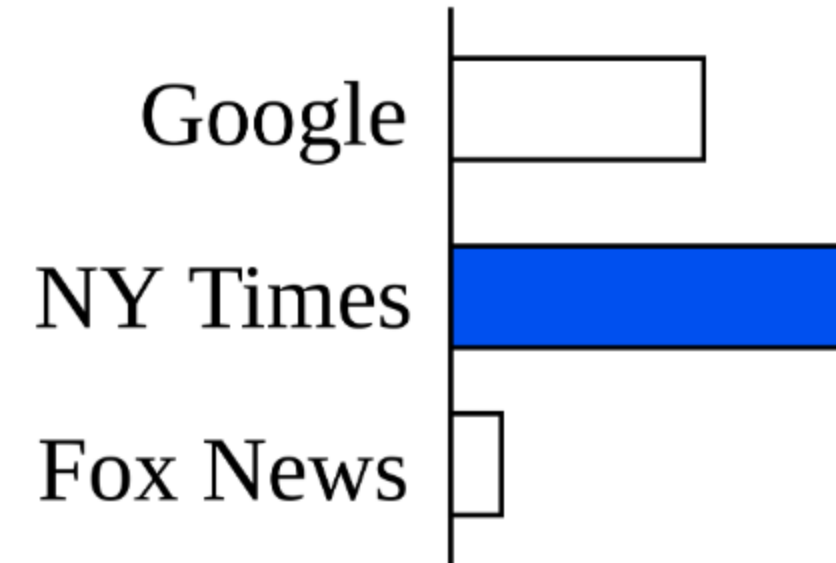
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- More data



Web Browsing for Political Polling

- Can website visits predict political leanings?
- Example - news websites
- More data
- Fully automated



Prior Work

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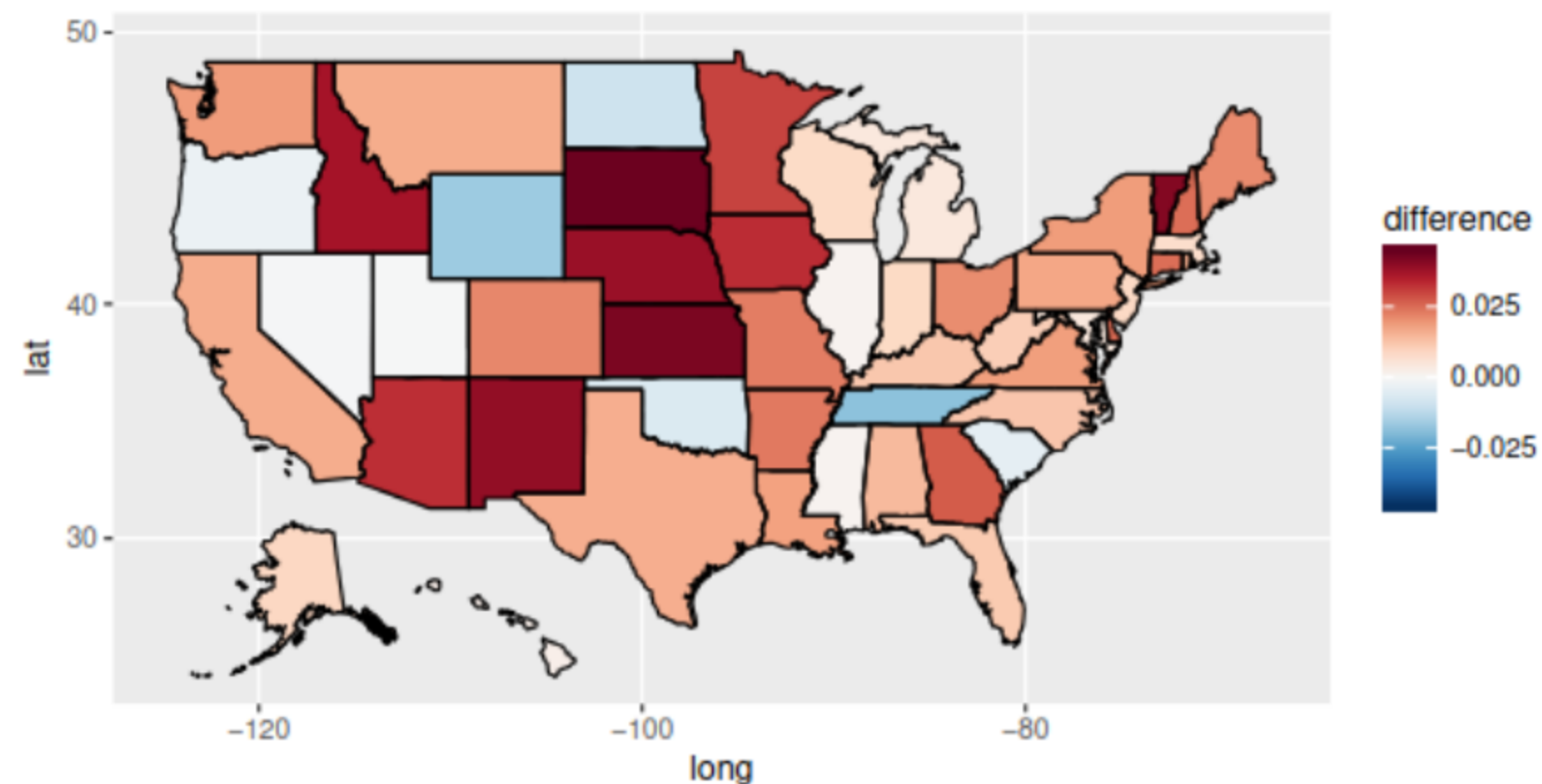


Figure 8: Impact of the 'Comey letter' at the state level.

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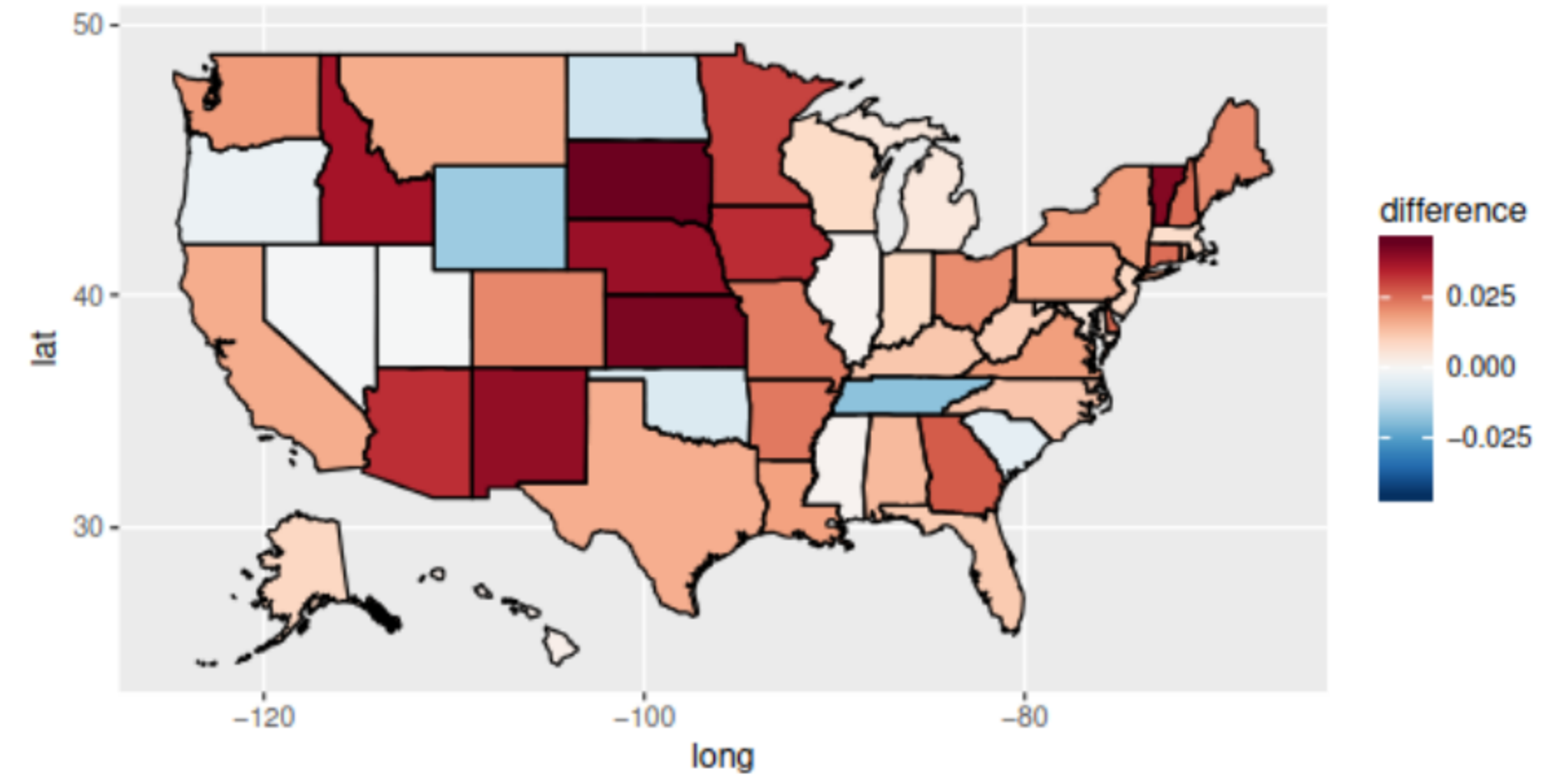
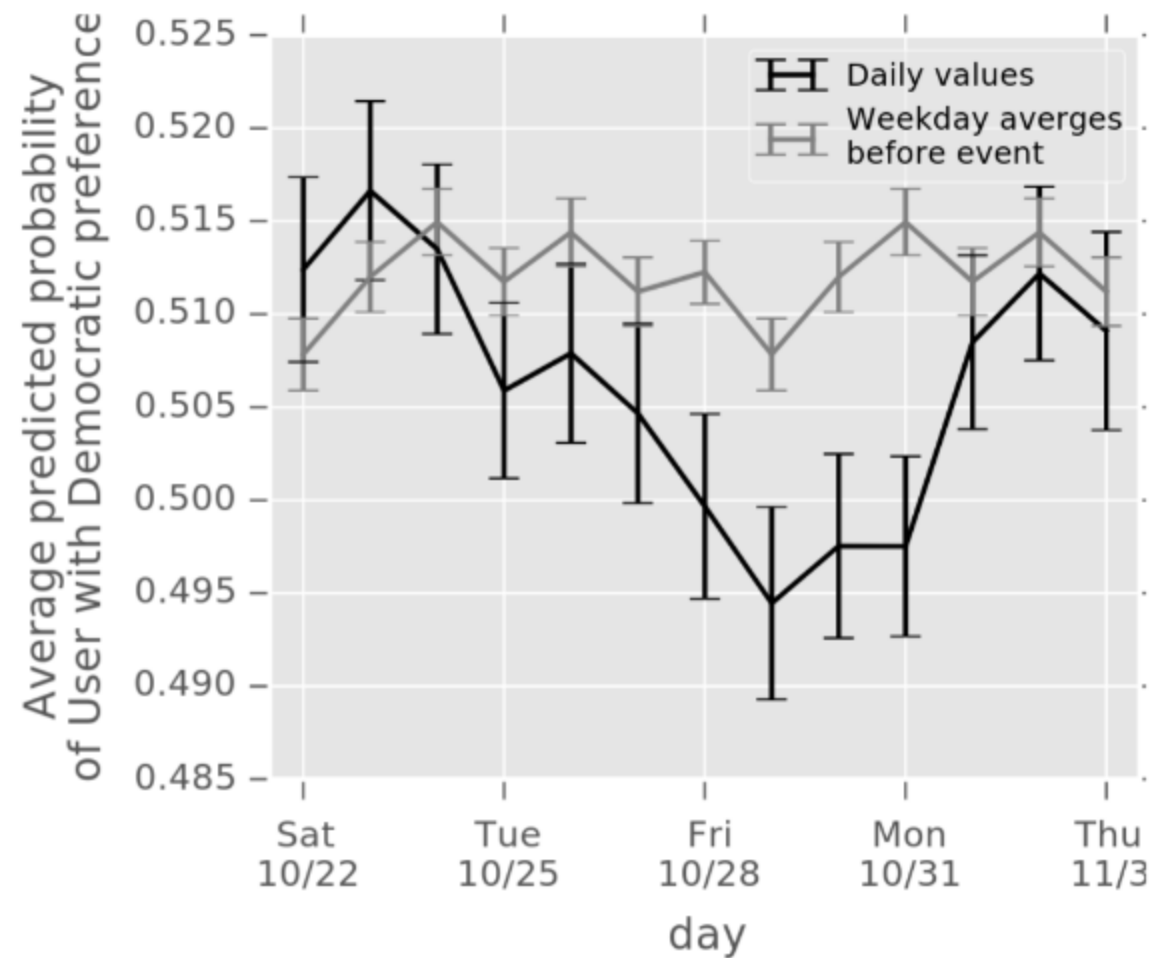


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Prior Work

- Web browsing behavior can predict voting results
- Quantifying the 'Corney letter' (Comarella et al.)
- Social media referrals are the best signal

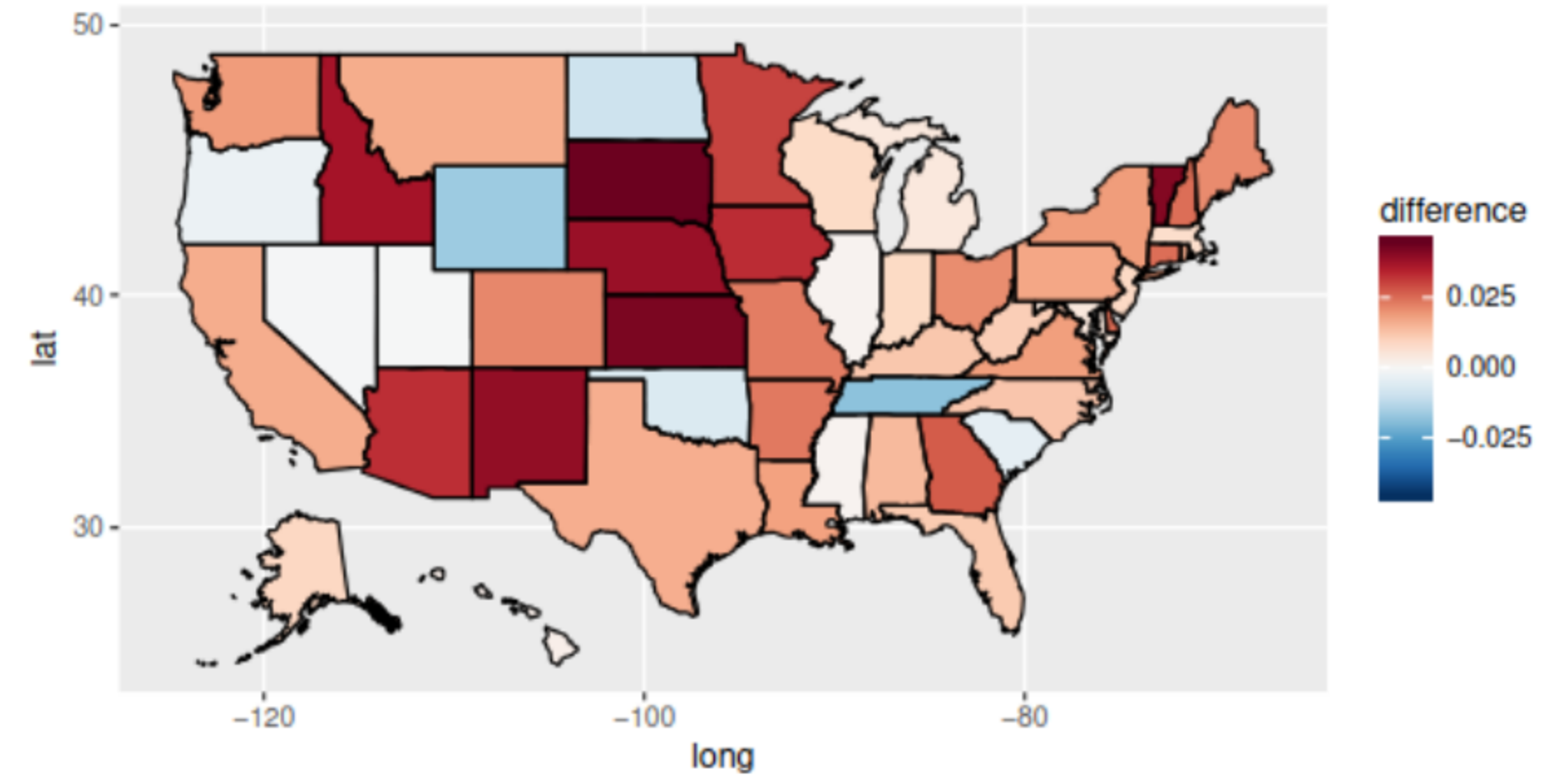
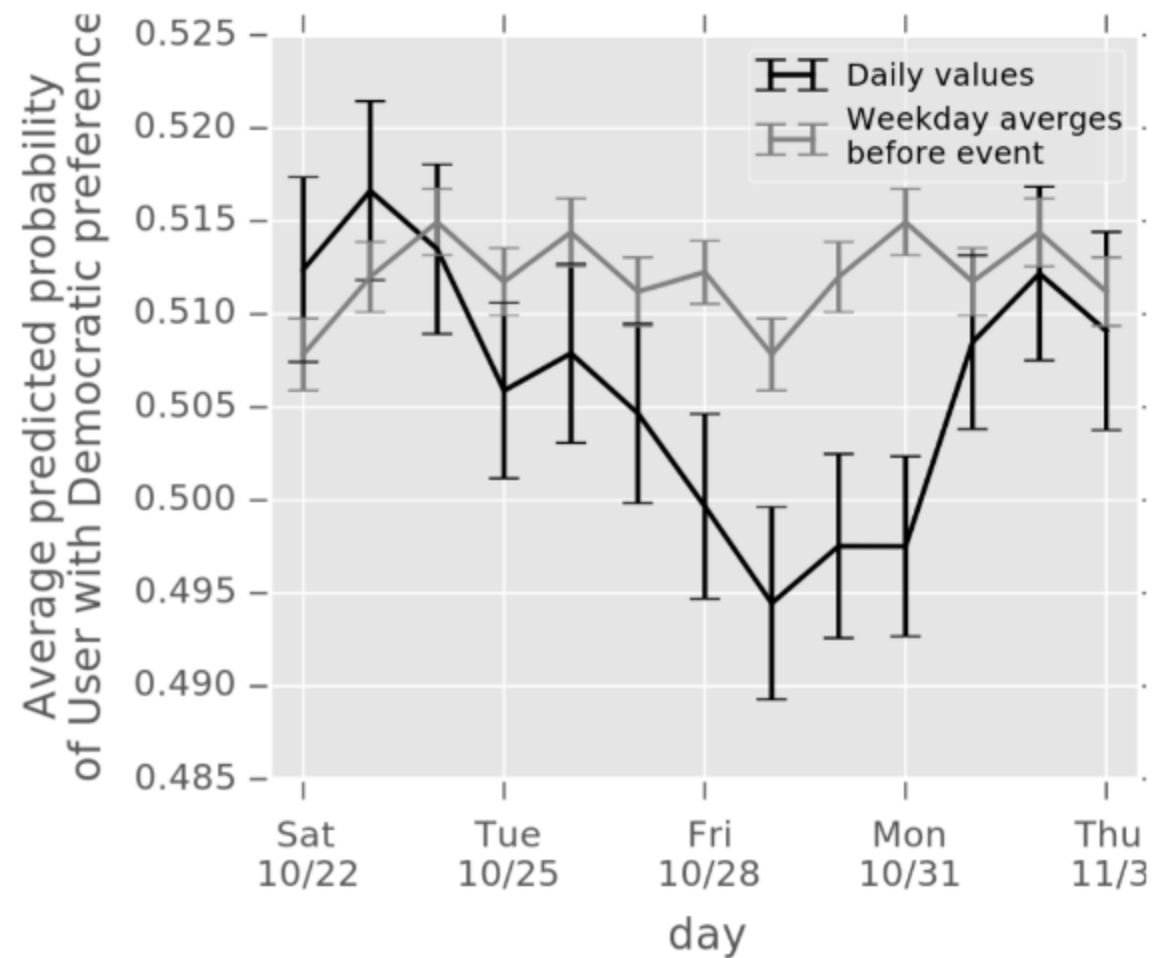


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Prior Work

- Web browsing behavior can predict voting results
- Quantifying the 'Corney letter' (Comarela et al.)
- Social media referrals are the best signal
- Used large plaintext dataset

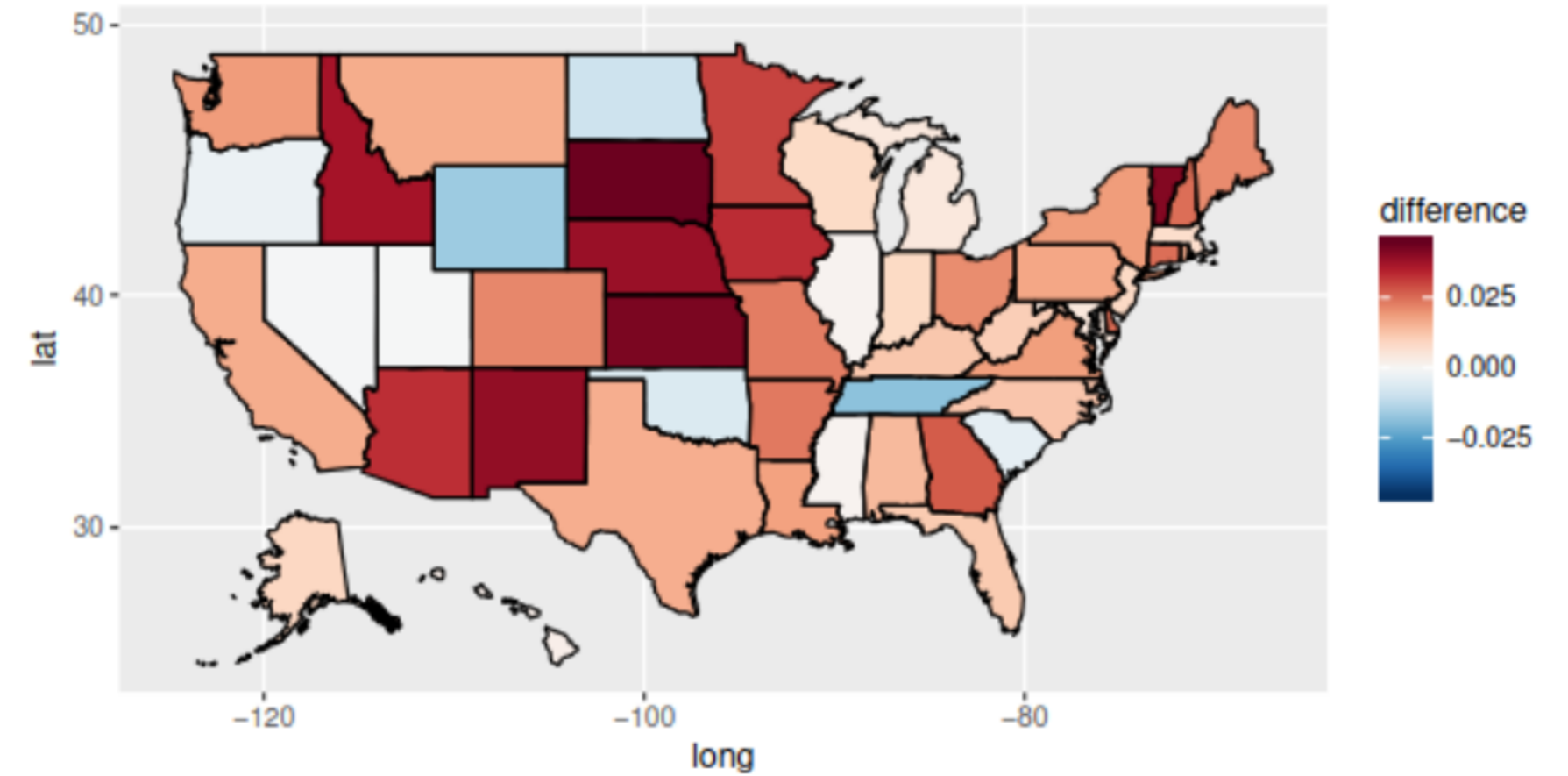
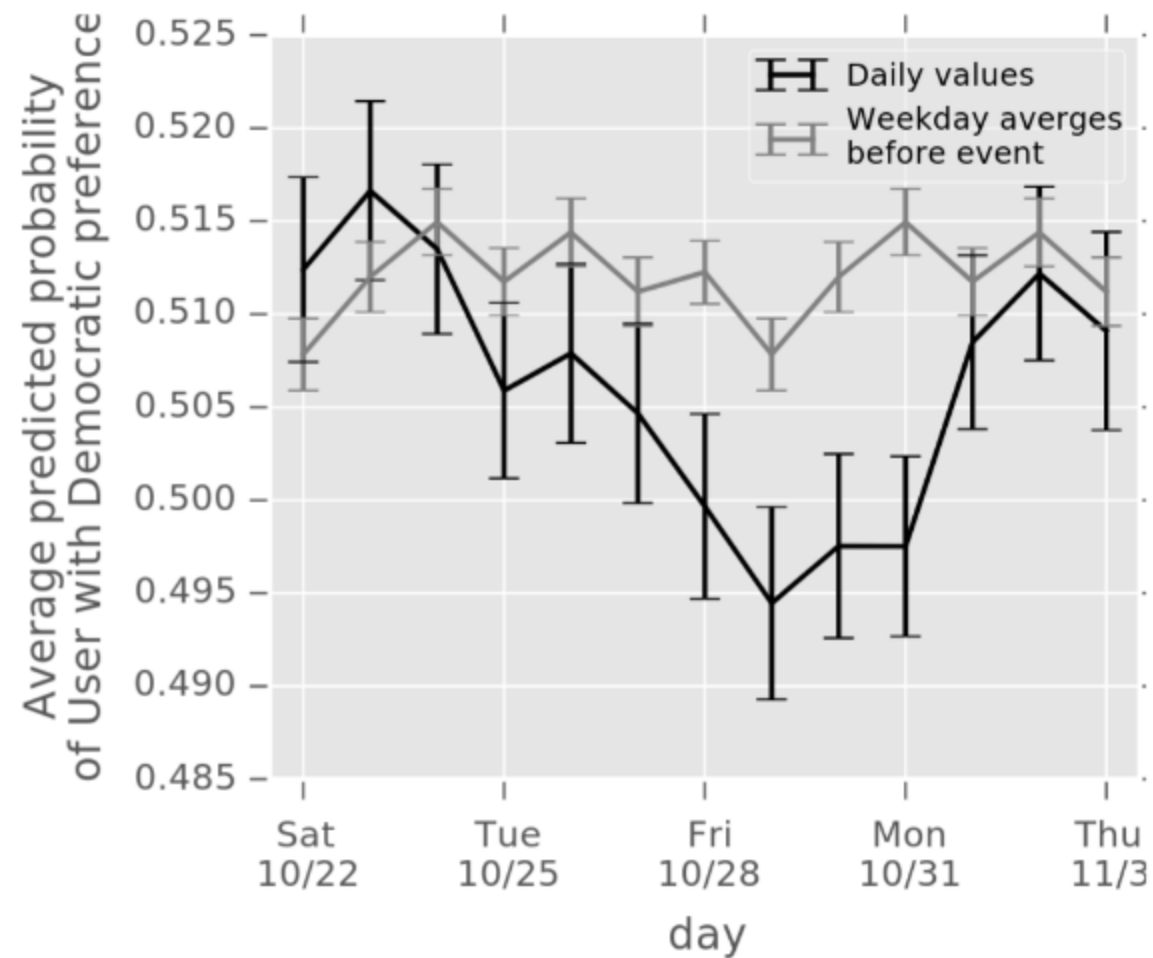


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Two Approaches to Political Polling

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Traditional Polling

Slow

Expensive

Coarse-grained insights

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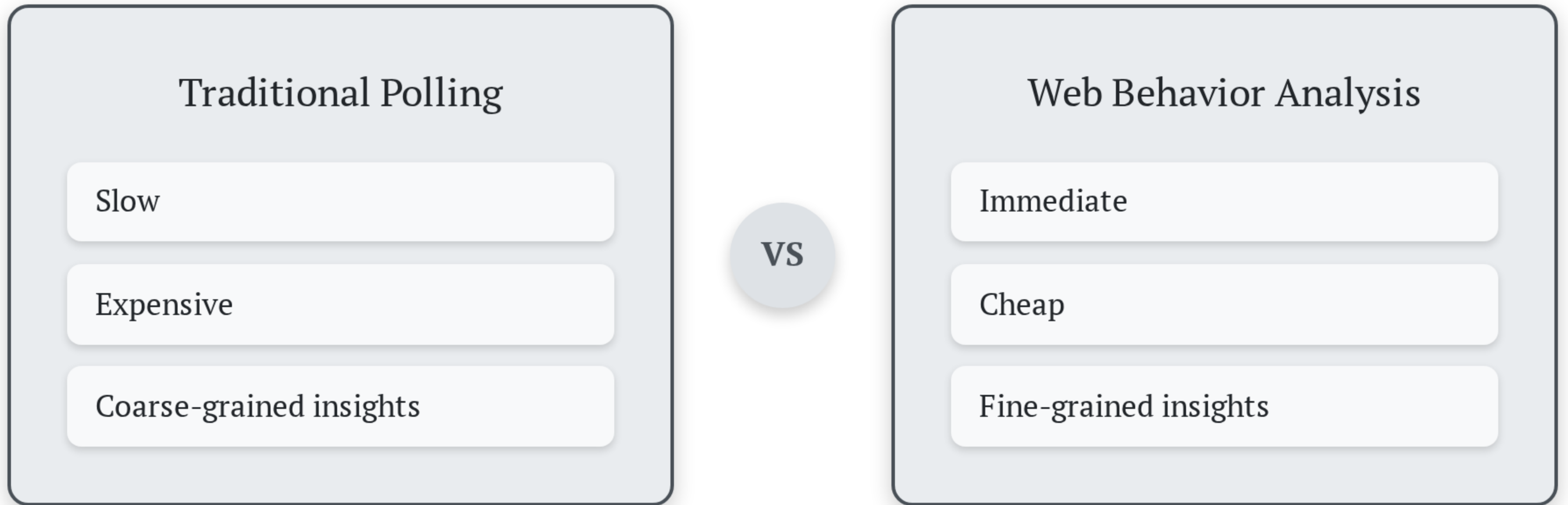
Web Behavior Analysis

Immediate

Cheap

Fine-grained insights

Two Approaches to Political Polling



What about privacy?

Our Contributions

Our Contributions

- We built a system for securely predicting political preferences from web browsing data

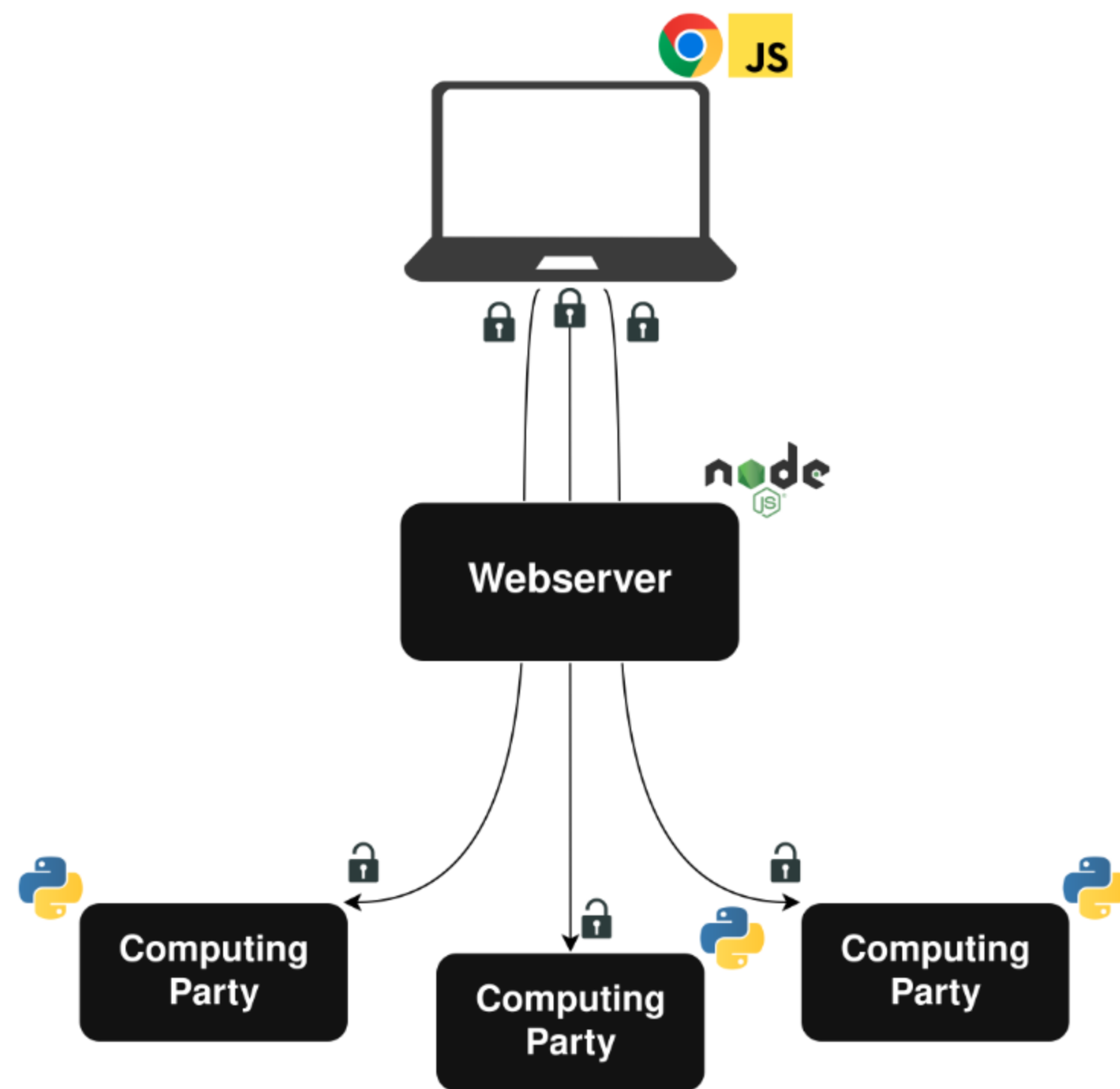
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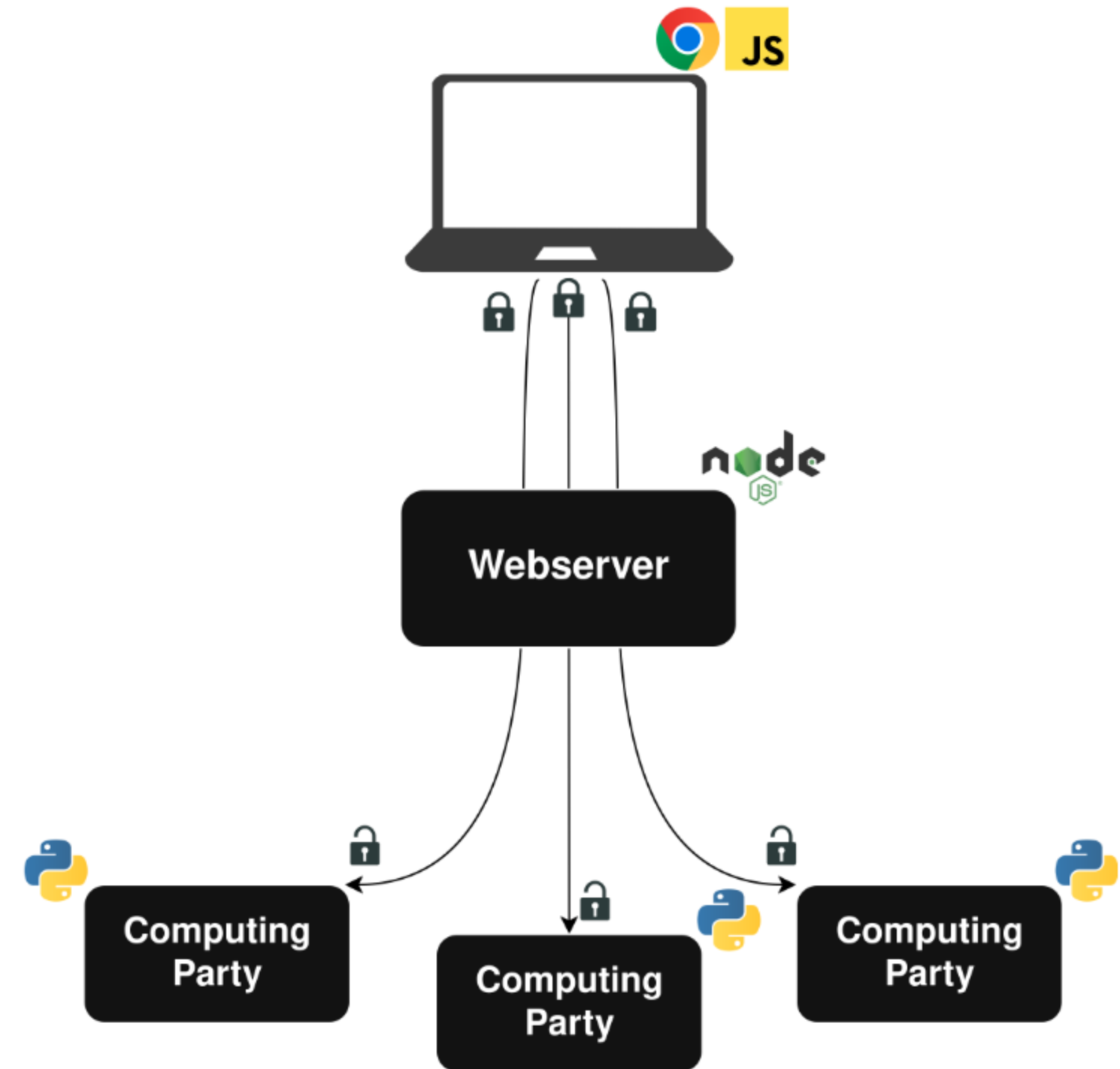
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- We collected and analyzed data from almost 8000 unique users
- All analysis took place under MPC

System Design



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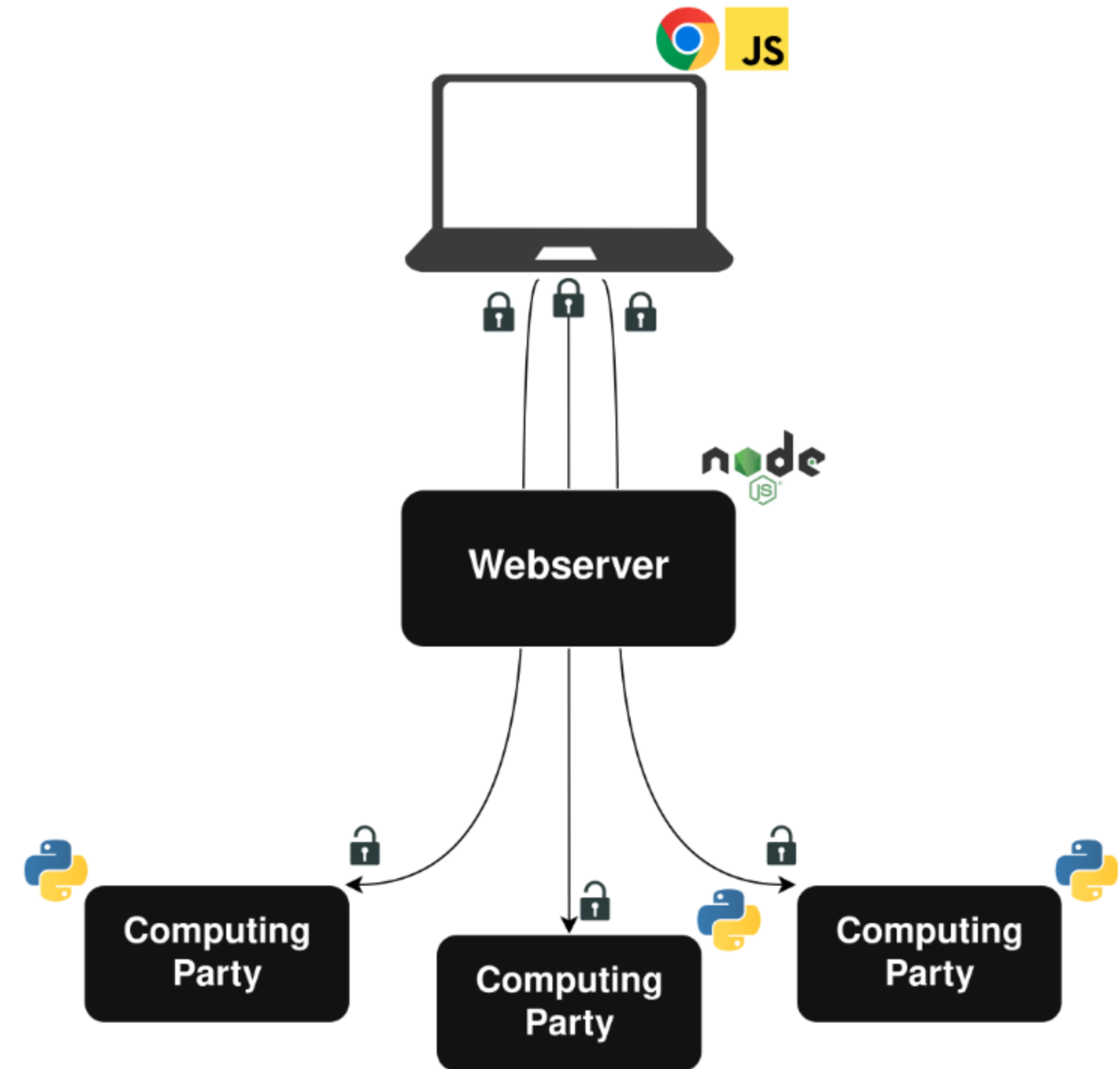
Client Plugin



System Design

Client Plugin

Webserver

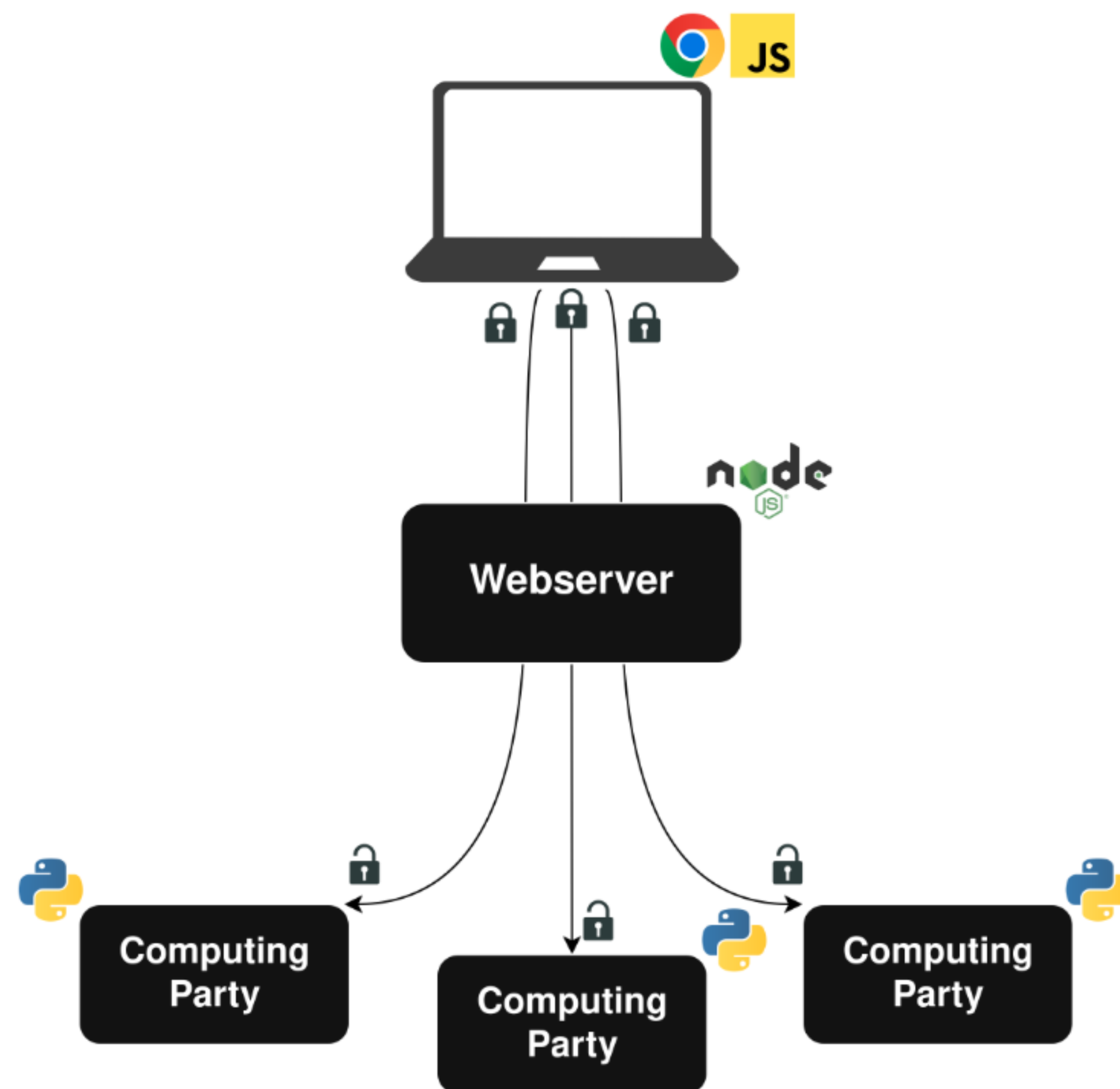


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MPC Backend

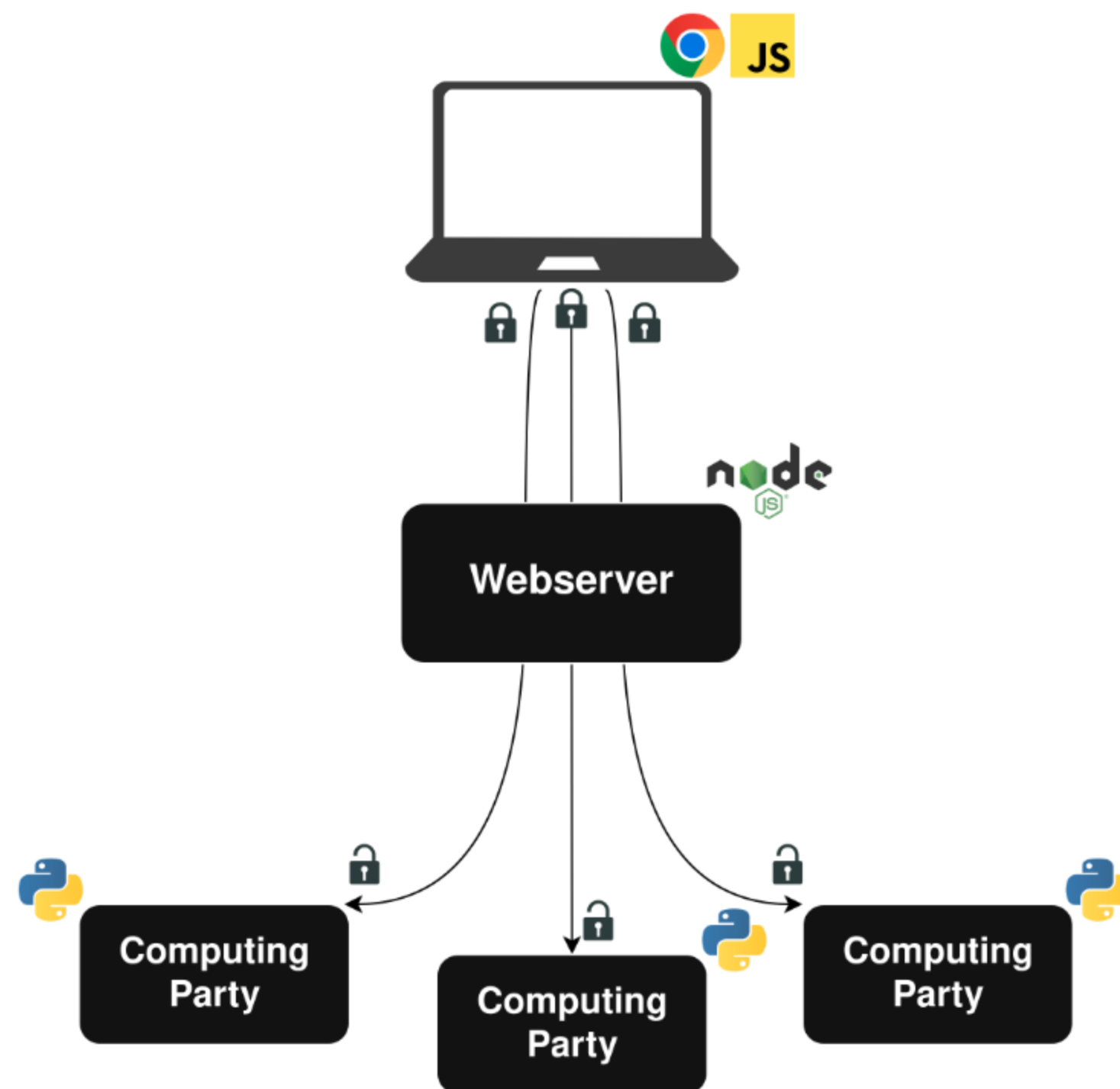


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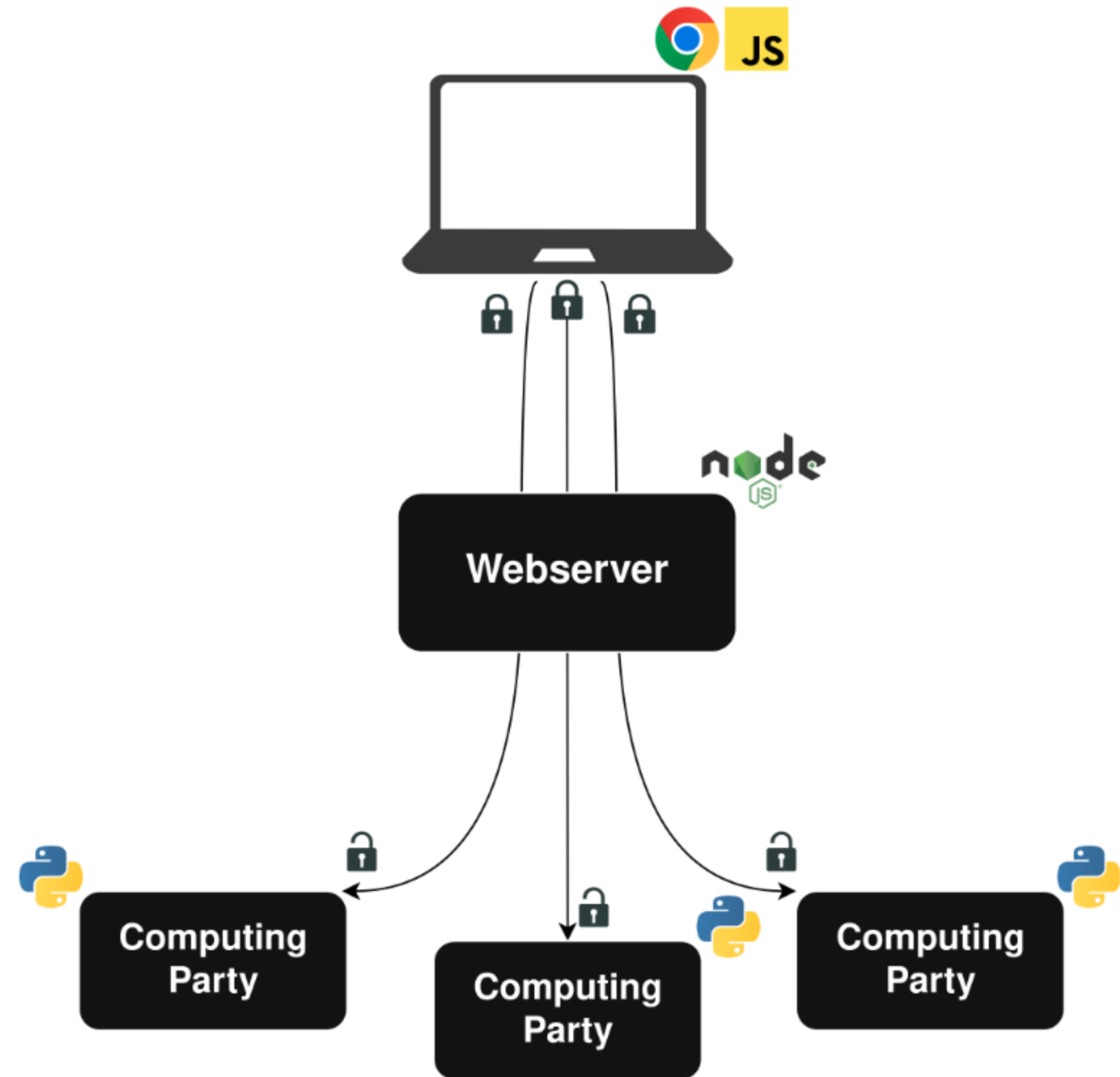
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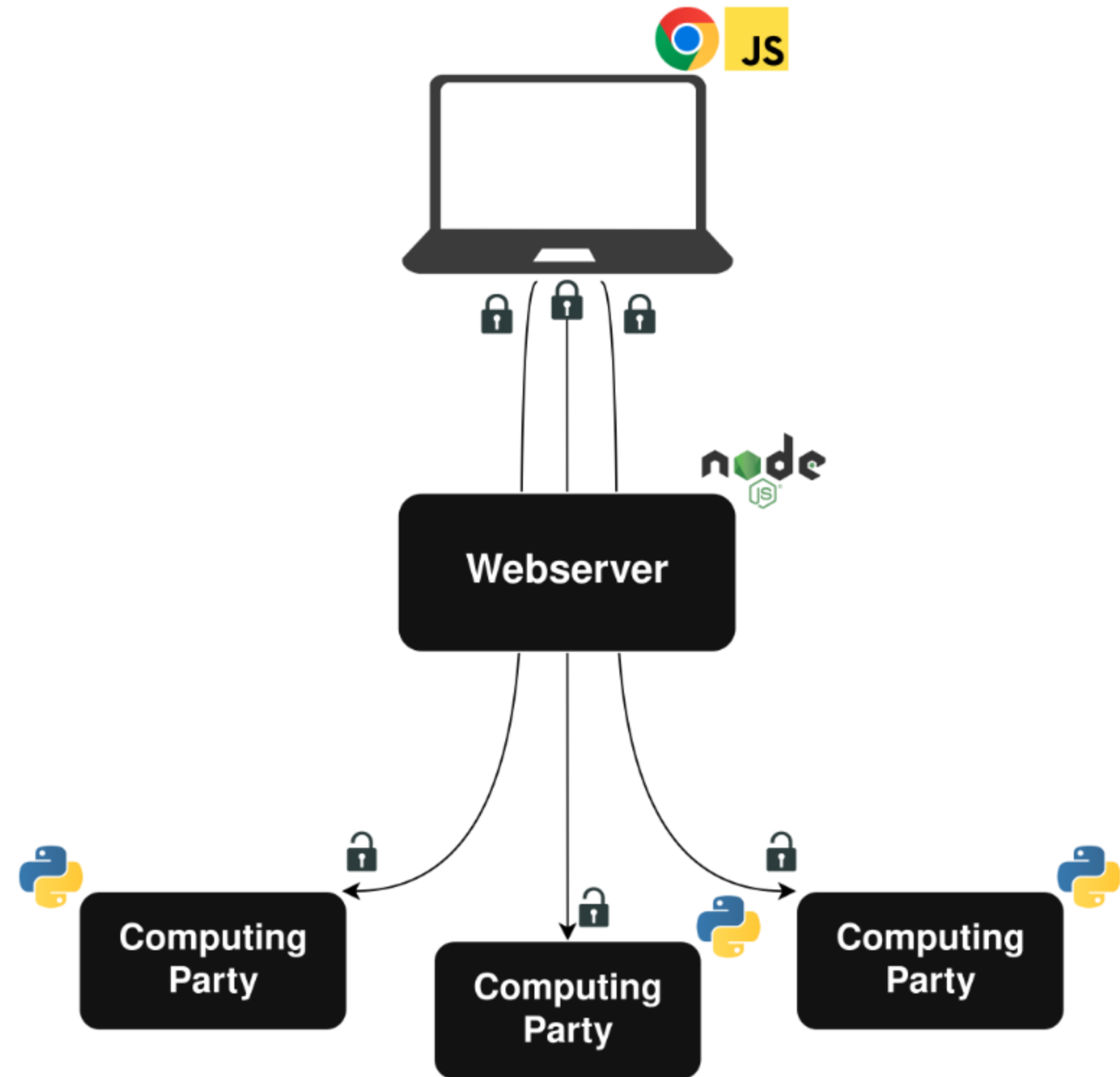


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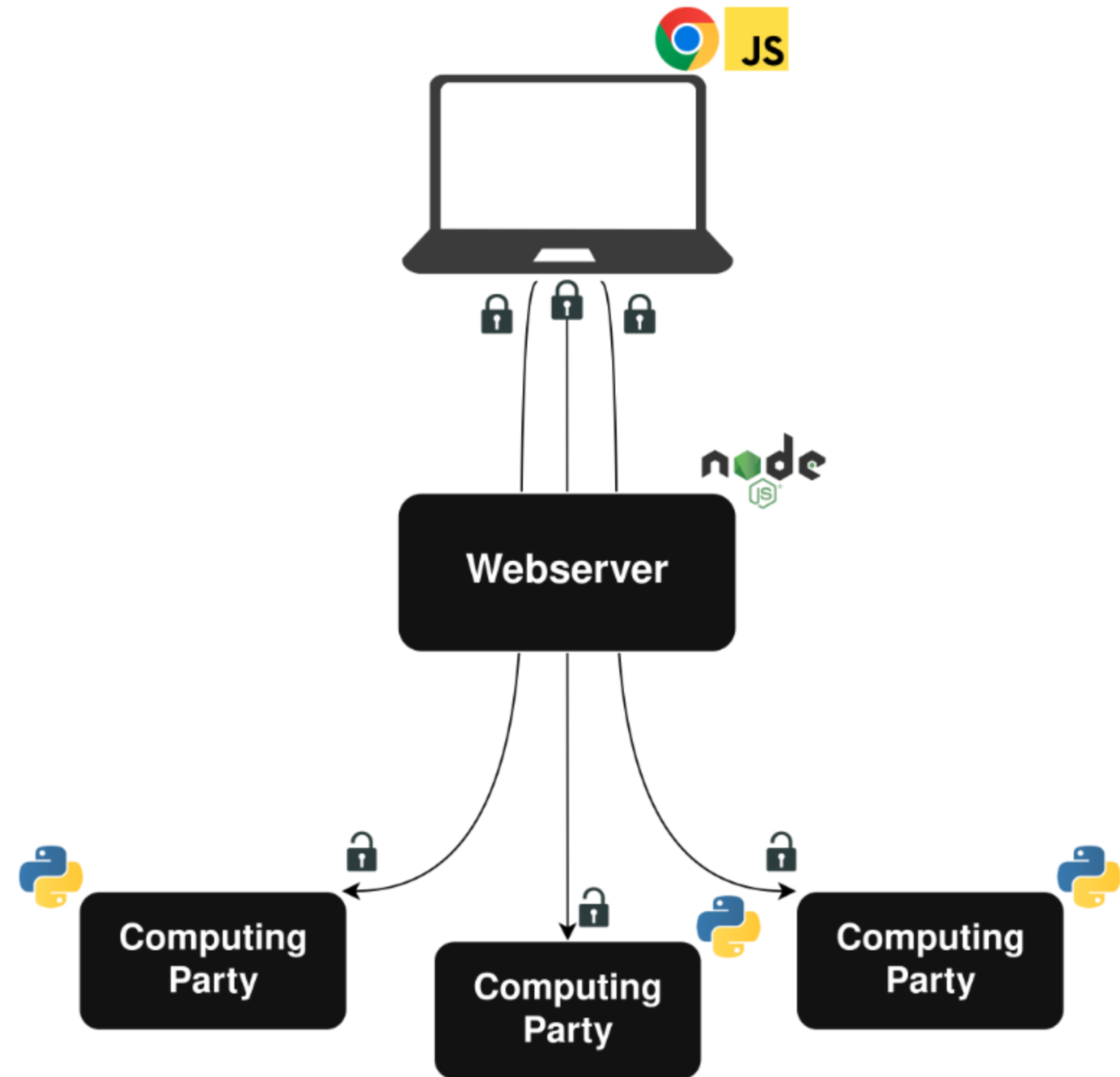
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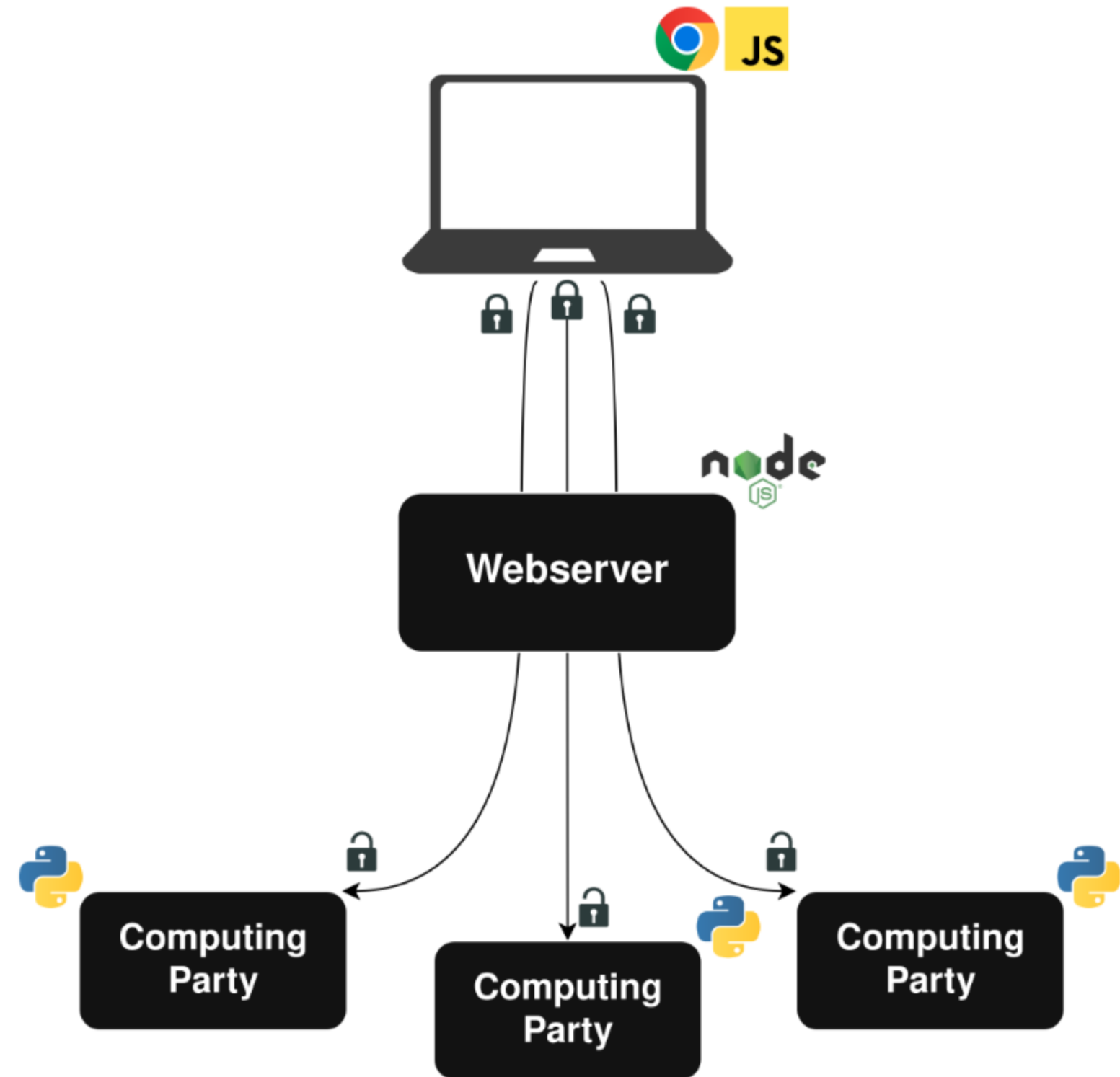
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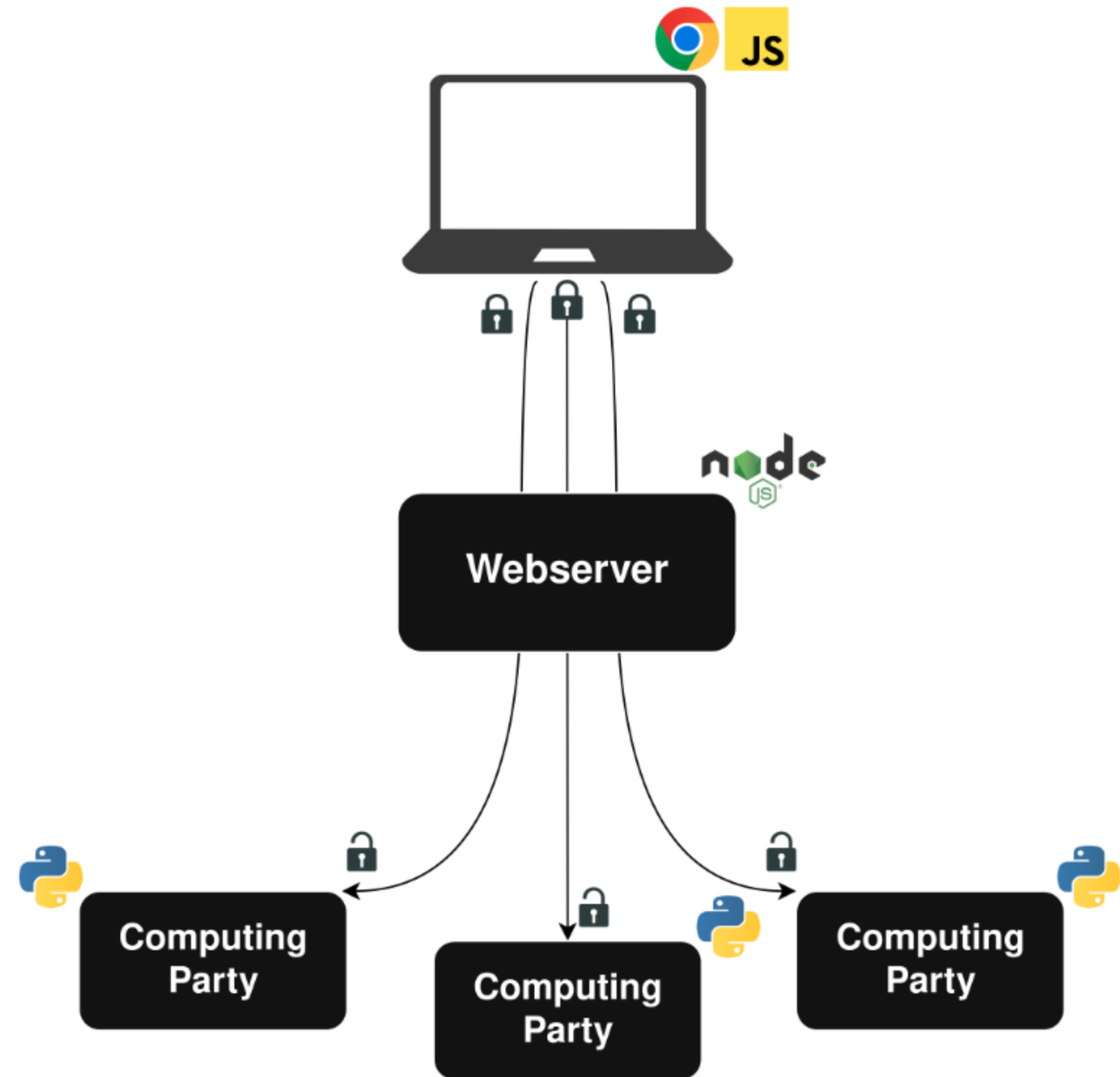
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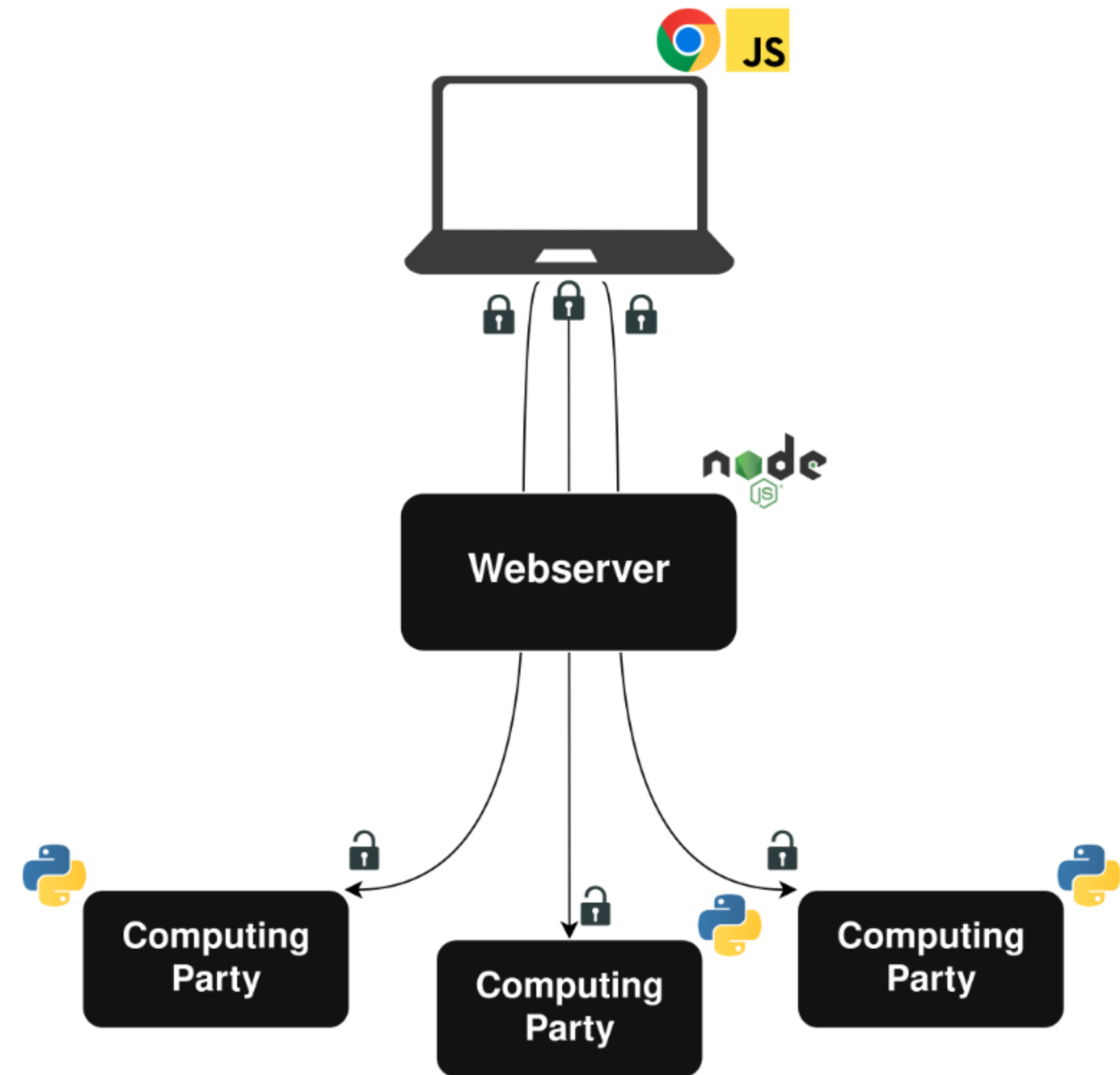


Client Plugin

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- Client-side secret sharing and encryption
- Implementation is open source

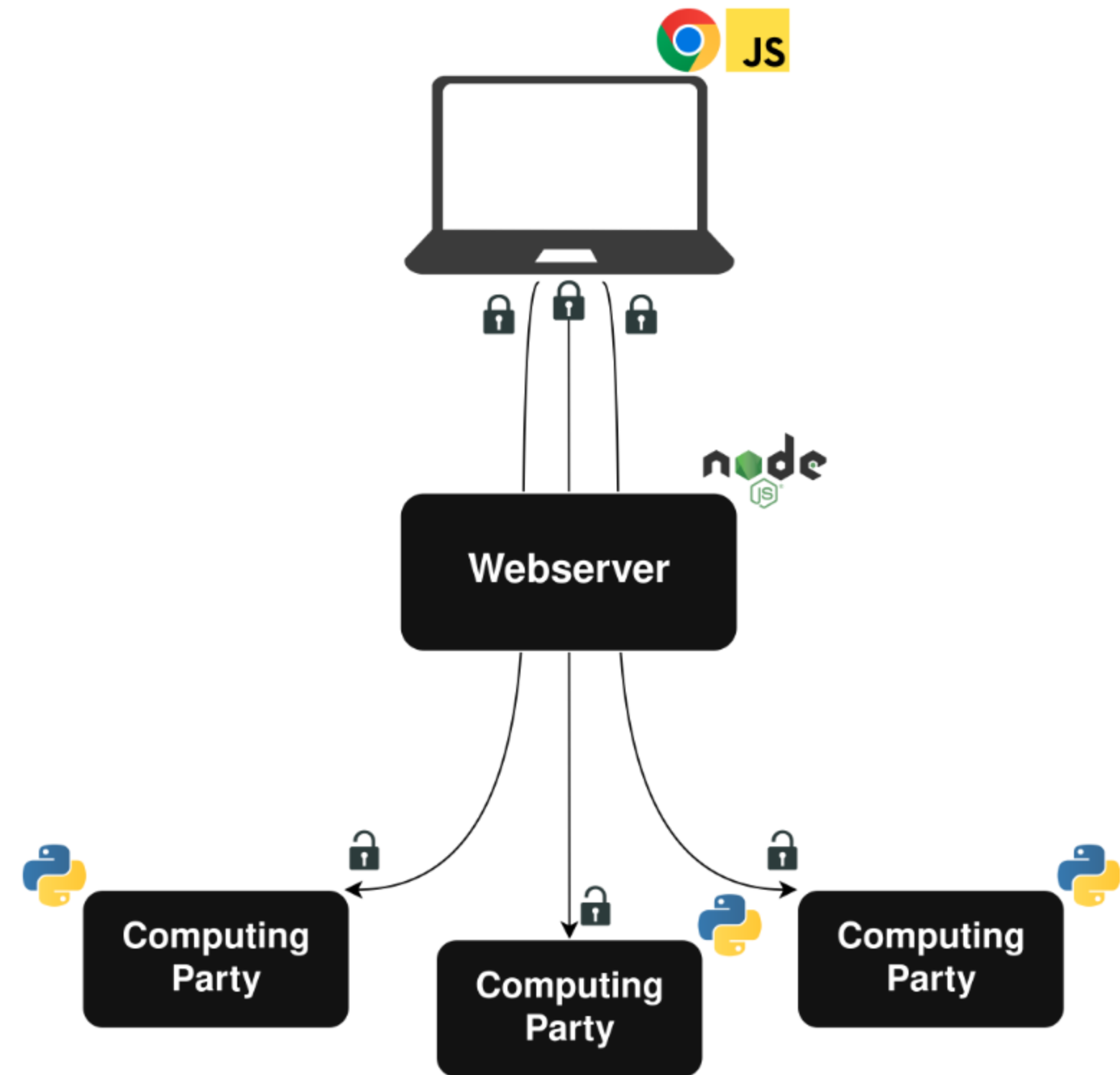


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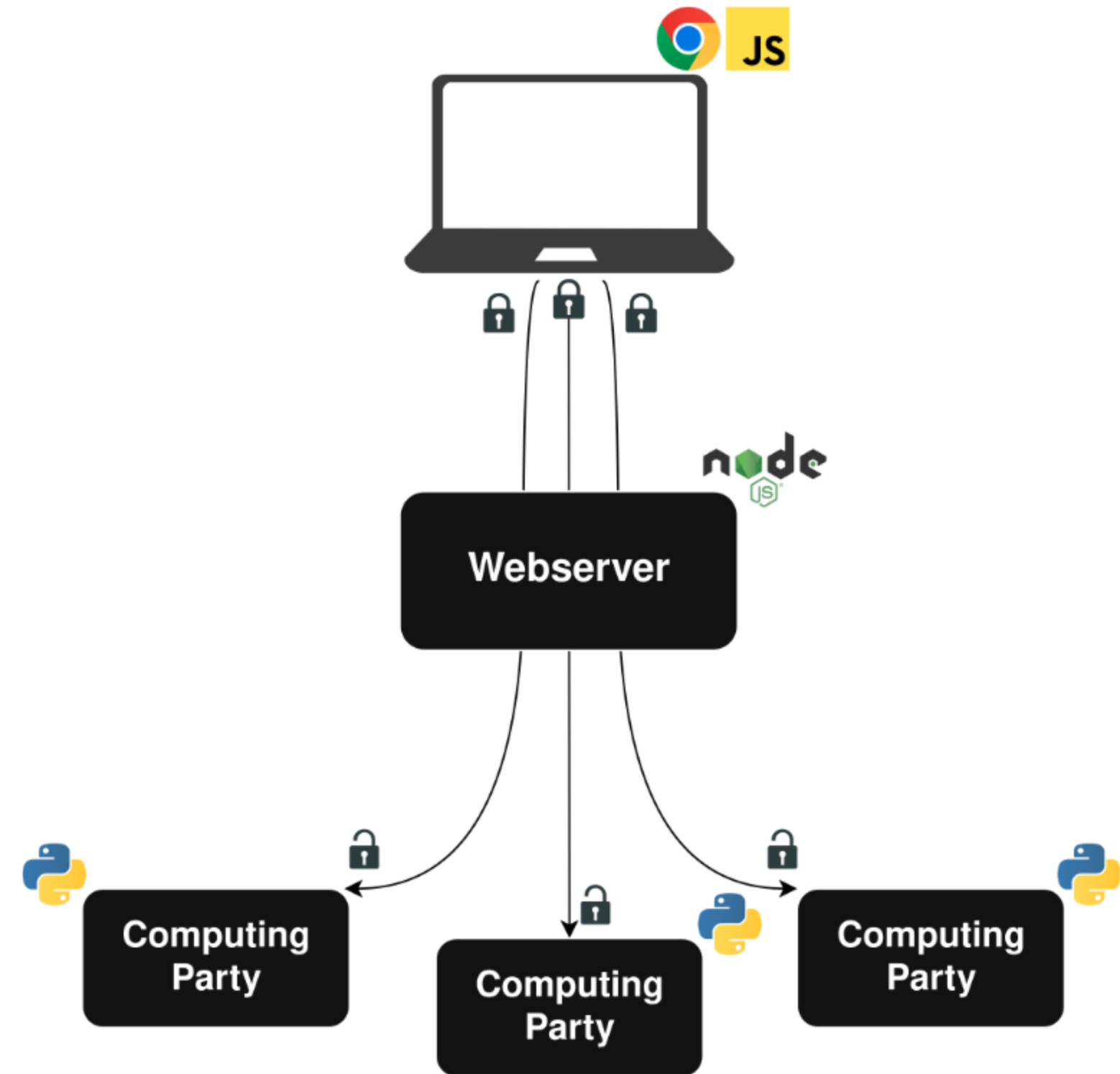
Webserver

- Simplifies interaction with clients



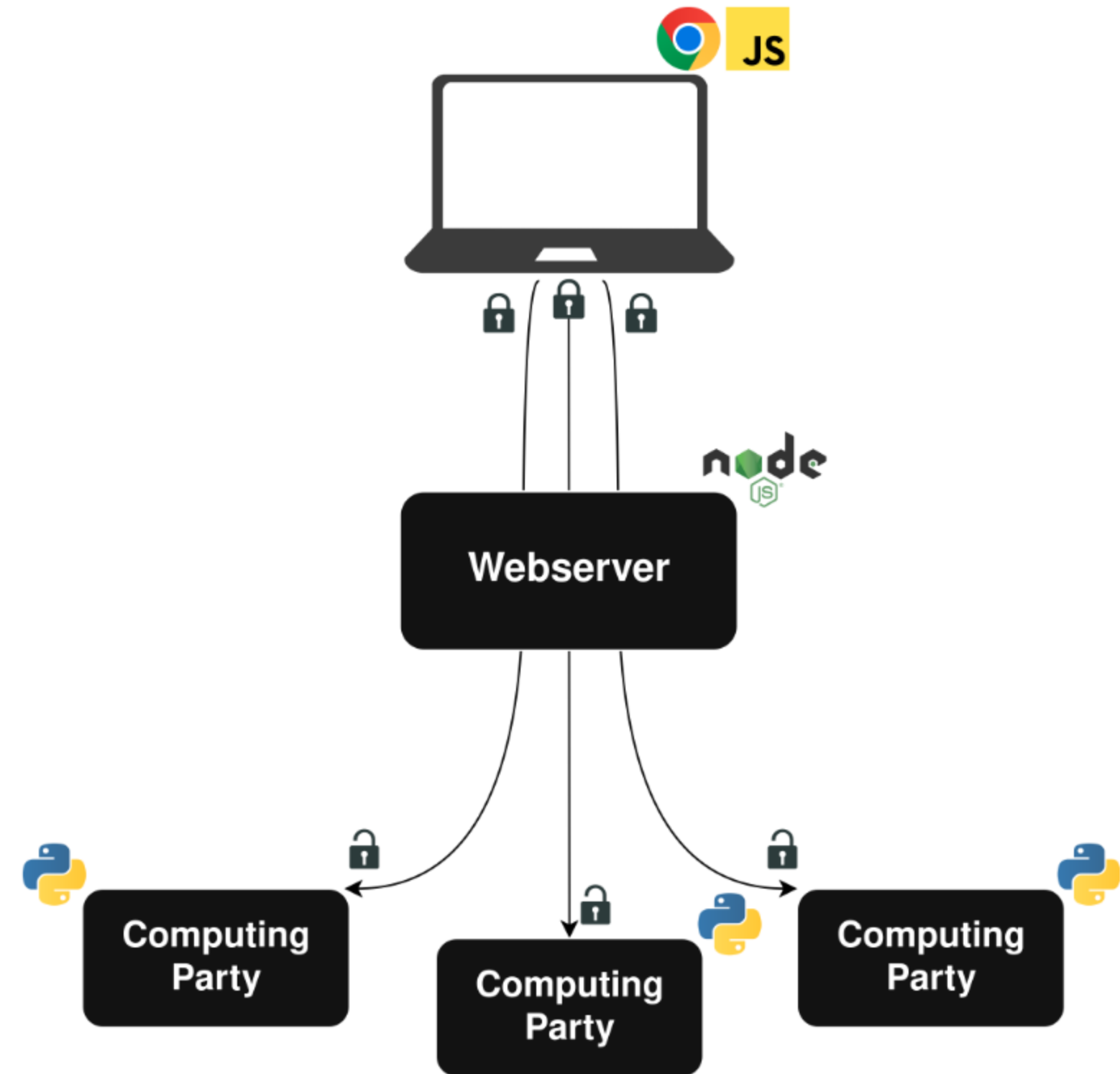
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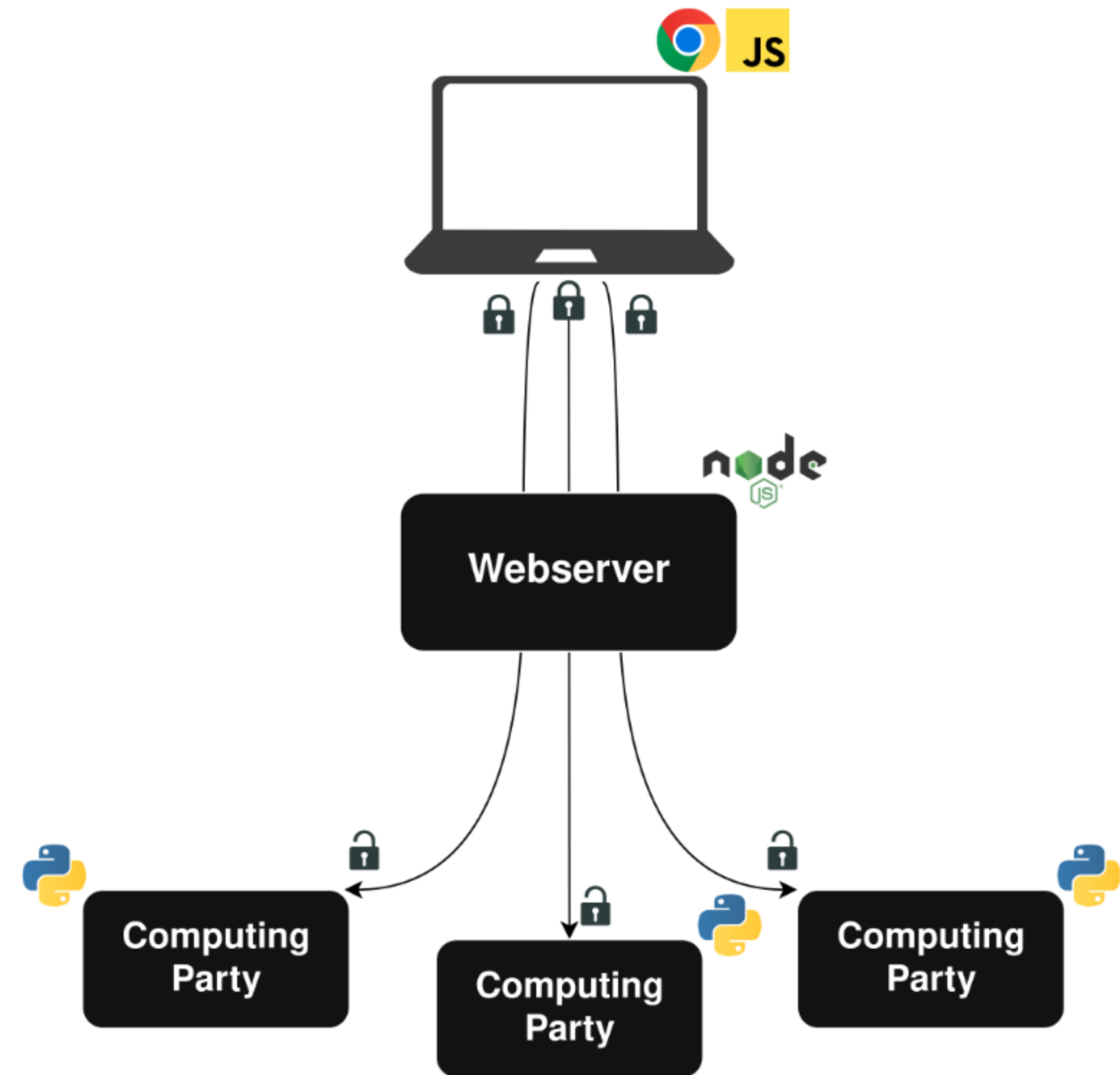


Webserver

- Simplifies interaction with clients
- Collects basic metadata
- Never sees any private data

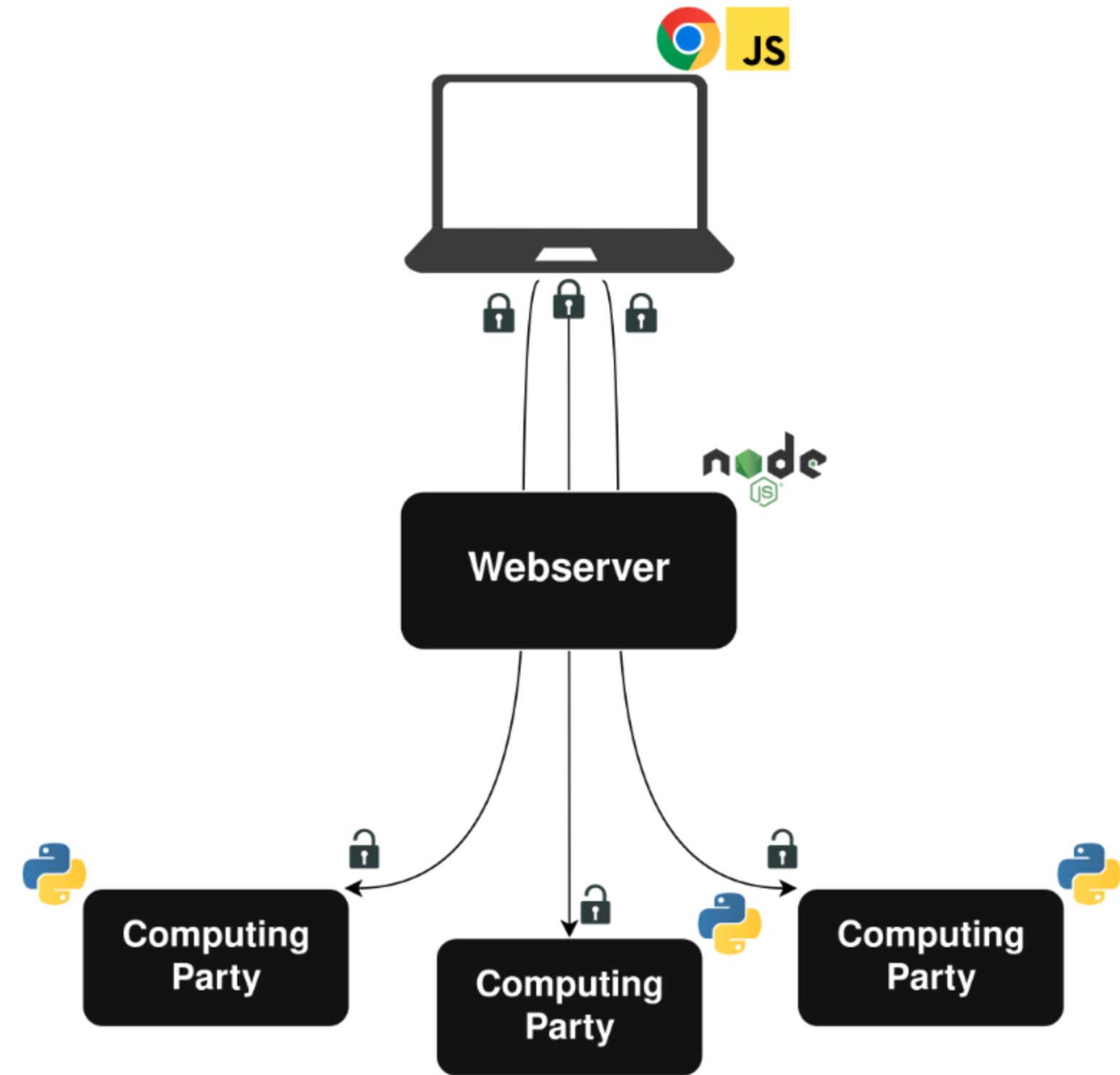


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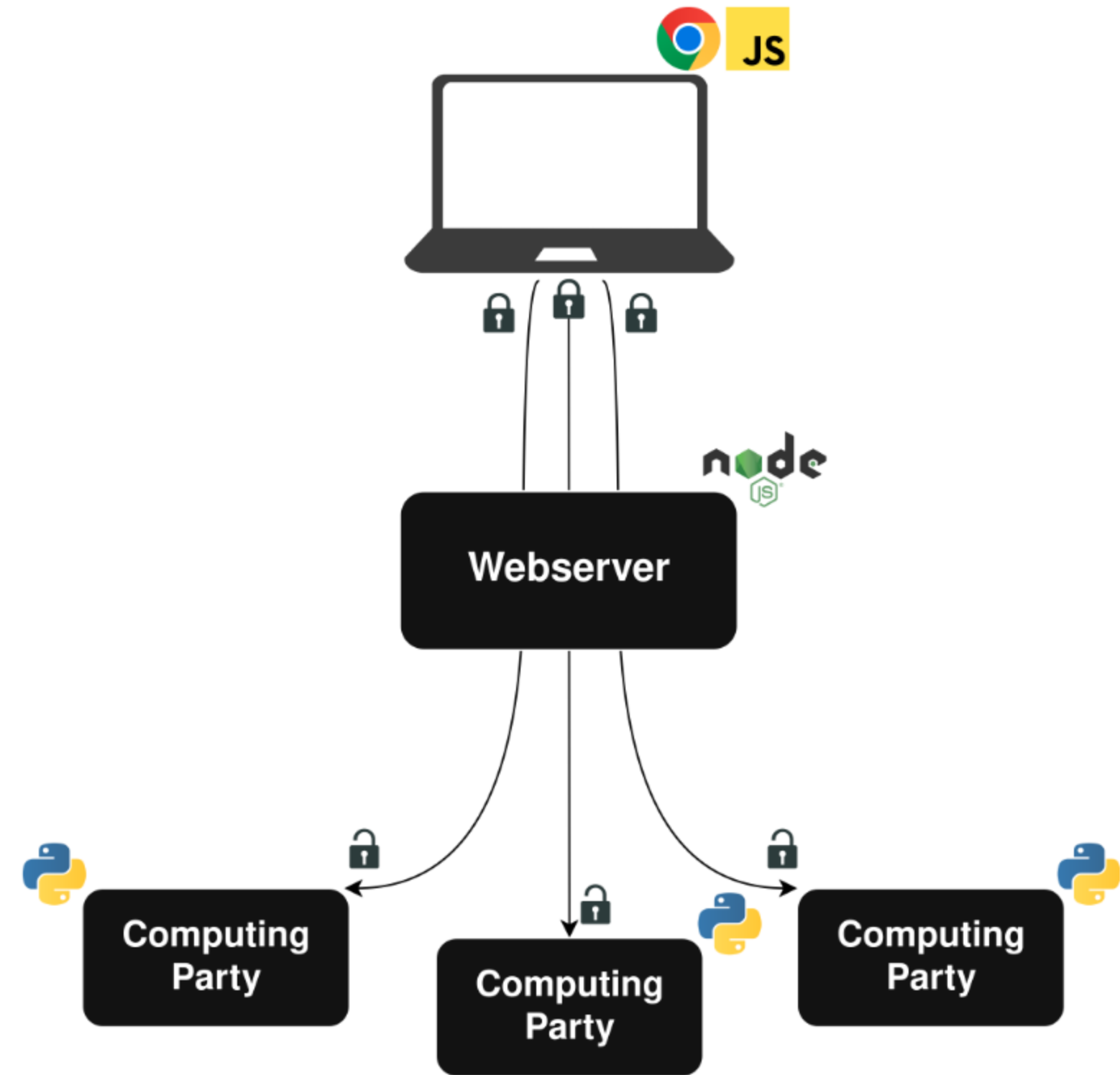
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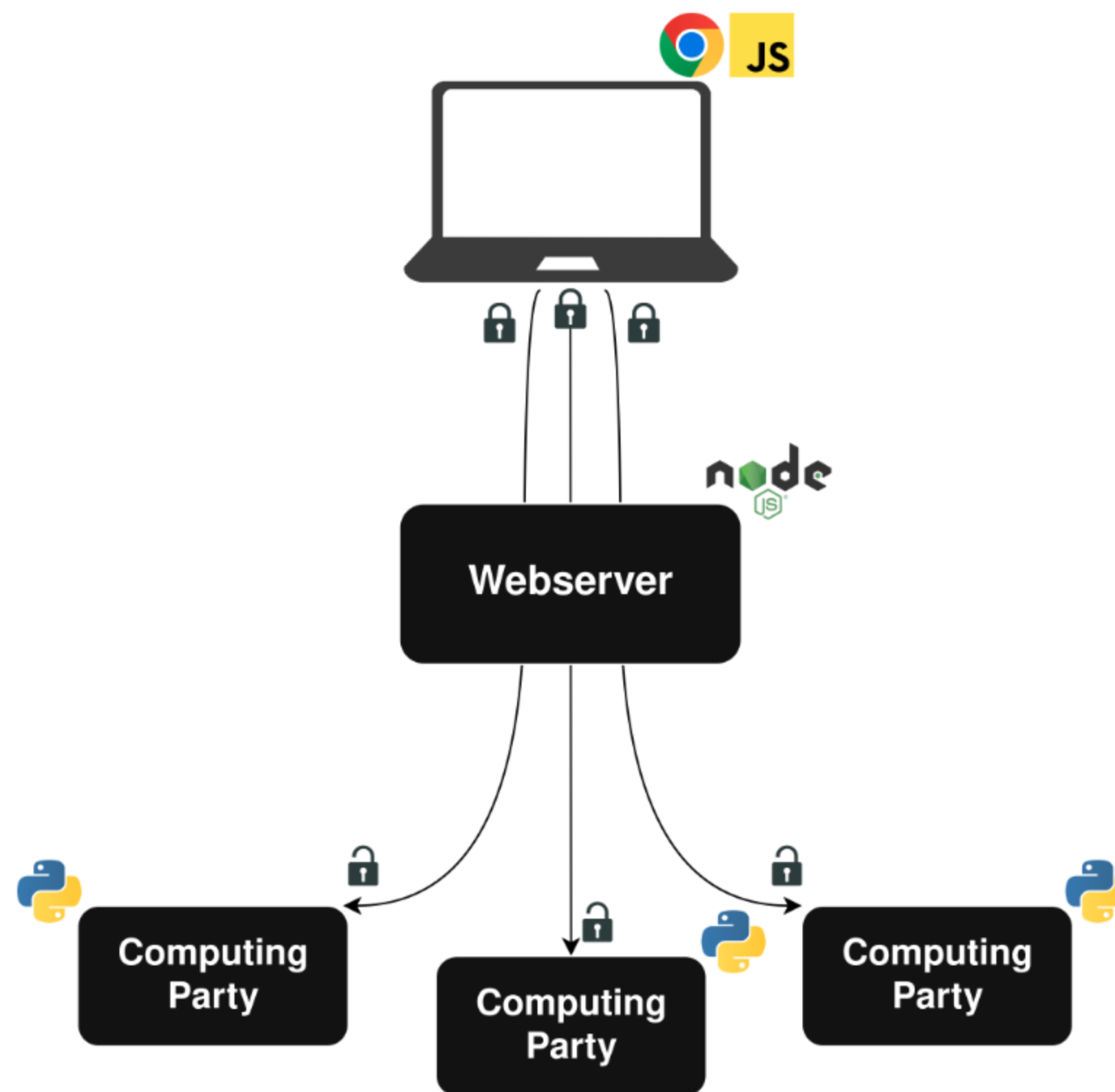
MPC Backend

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- We implemented an algorithm for LLP under MPC



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- The first implementation under MPC of an LLP model
- Mix of out-of-the-box components and custom implementations

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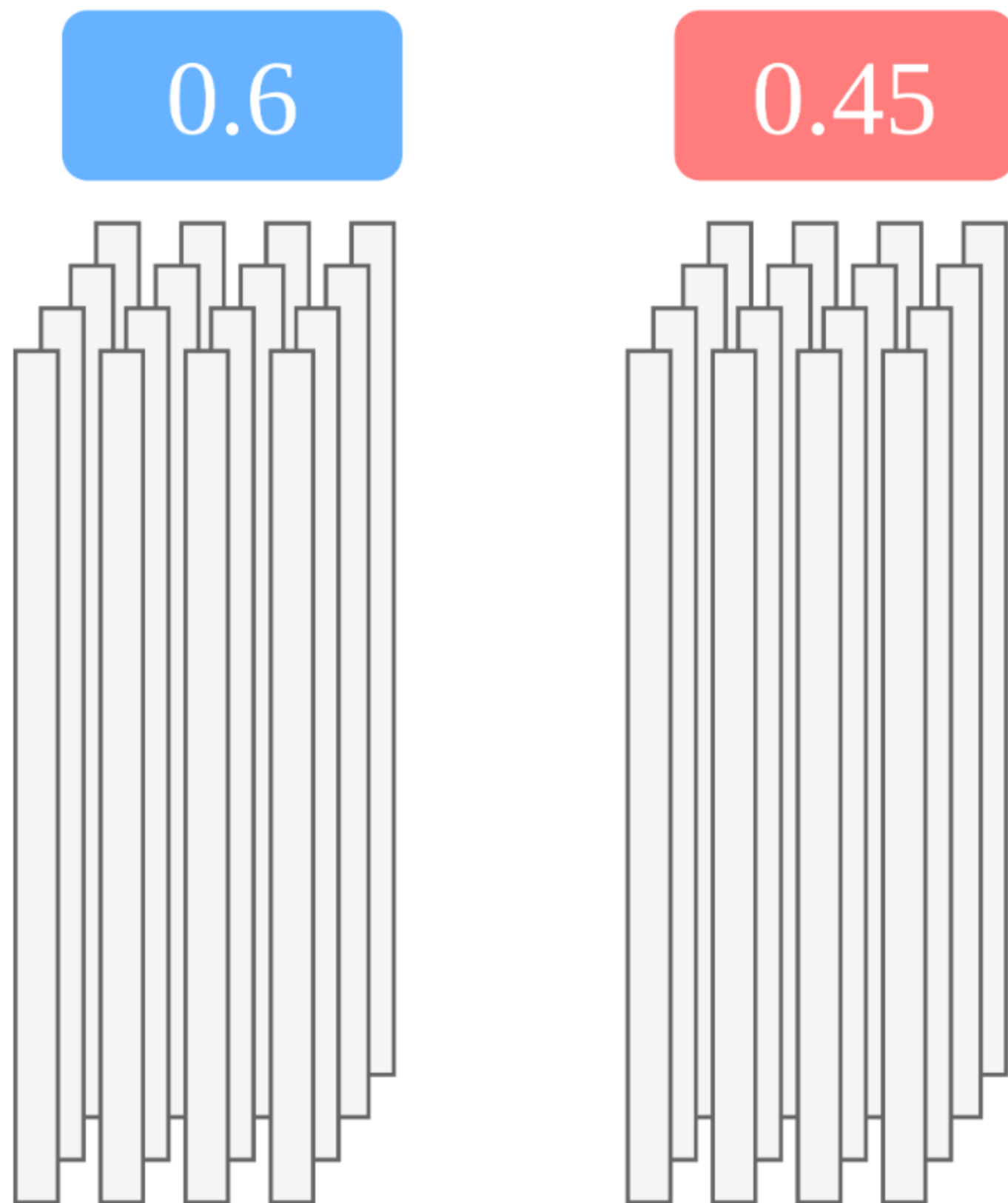
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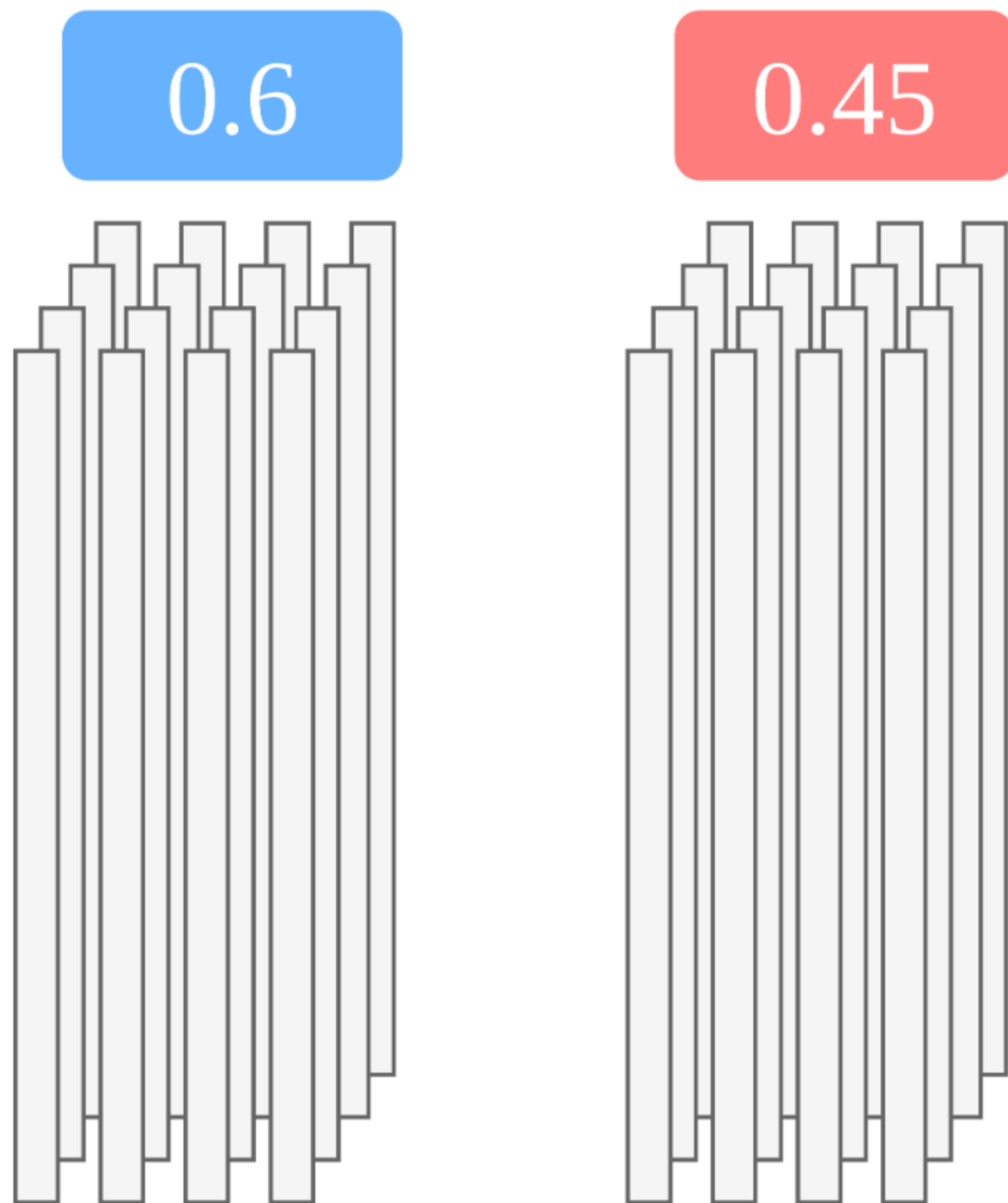
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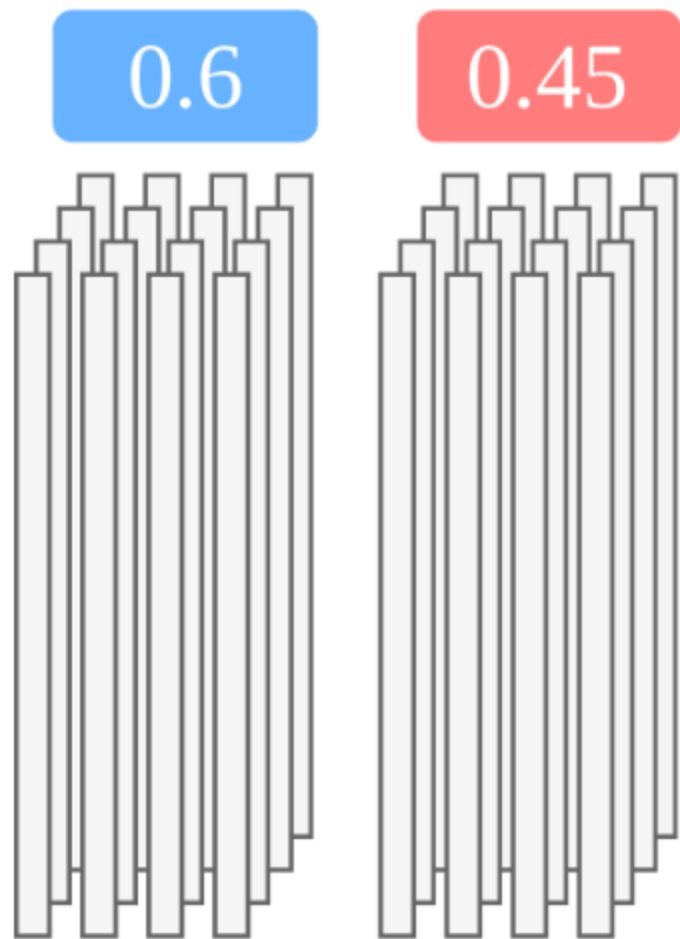
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- Train on aggregate ground truth

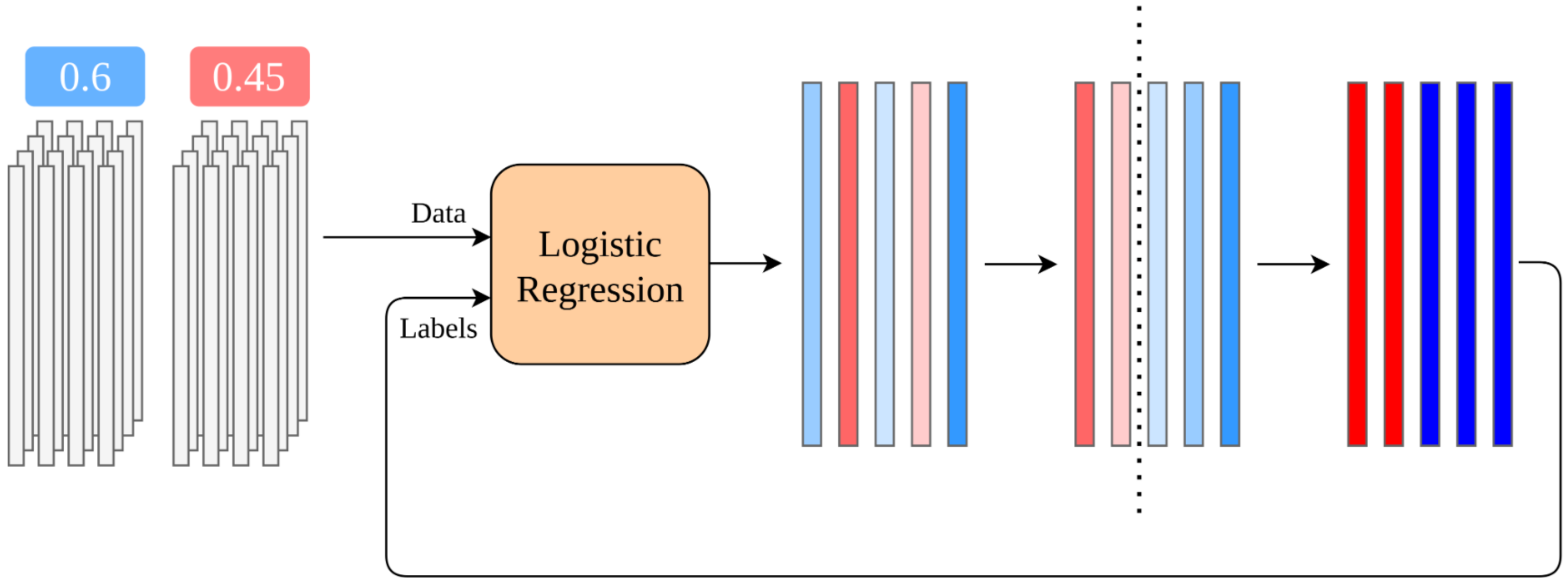


The Plaintext Algorithm

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Repeat Until Convergence

Translating to MPC

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initialize_labels()

repeat until convergence:
    model = train(data, labels)
    predictions = inference(model, data)

    for each state:
        compute_threshold(truth[state], predictions[state])

    for each user:
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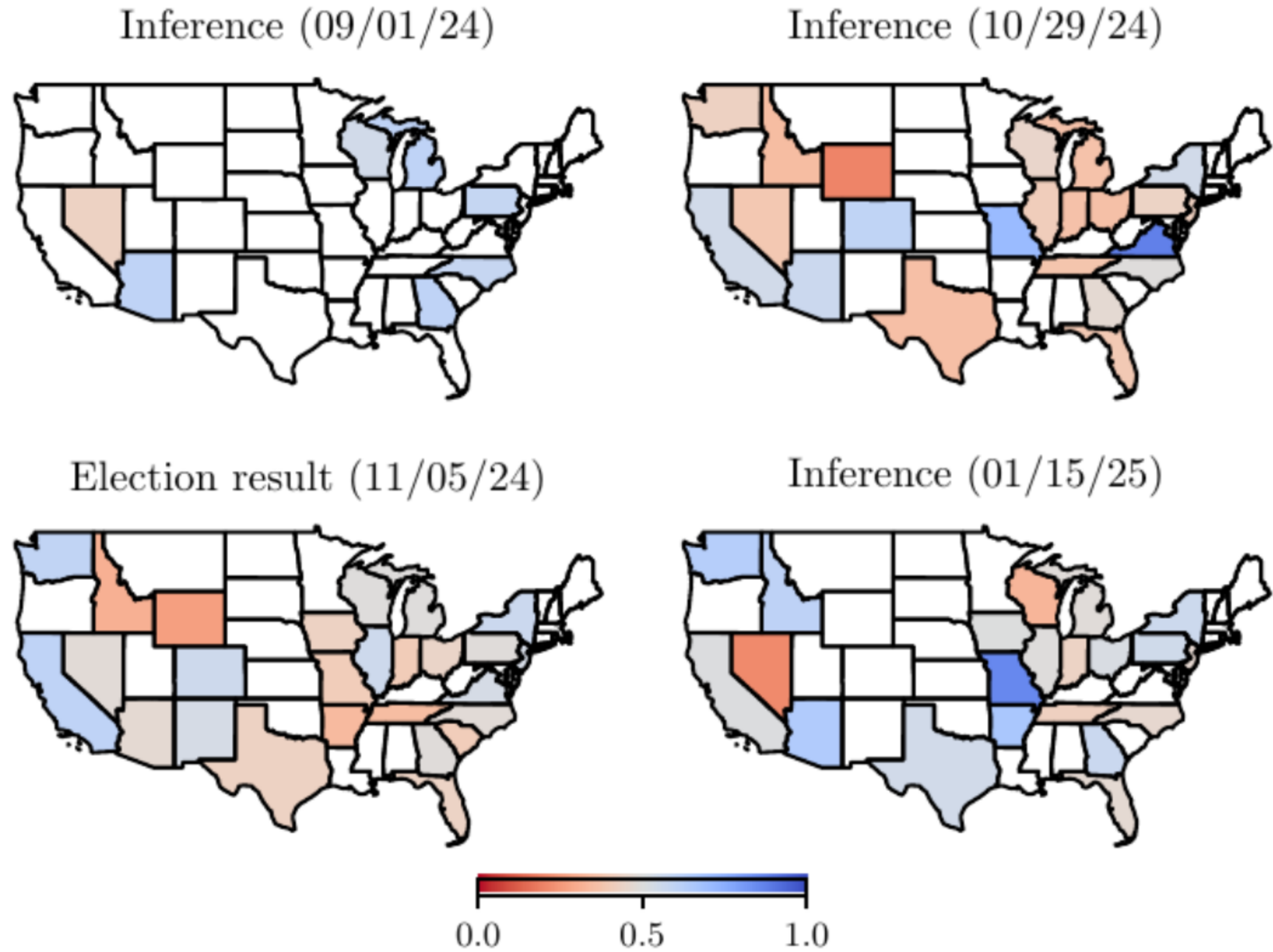
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Convergence Check

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- Sum the comparison results
- Compare sum with convergence threshold

Preliminary Results



Lessons Learned and Future Directions

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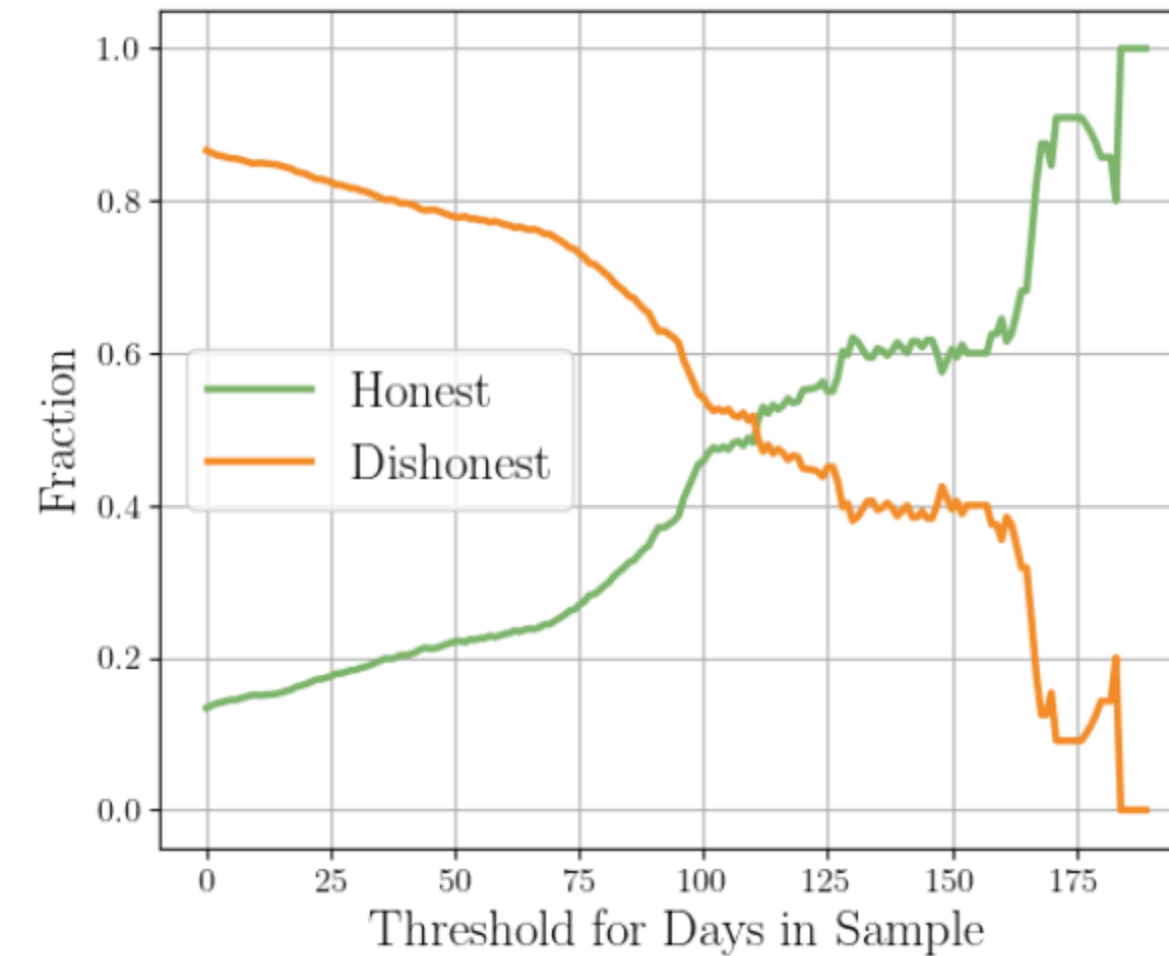
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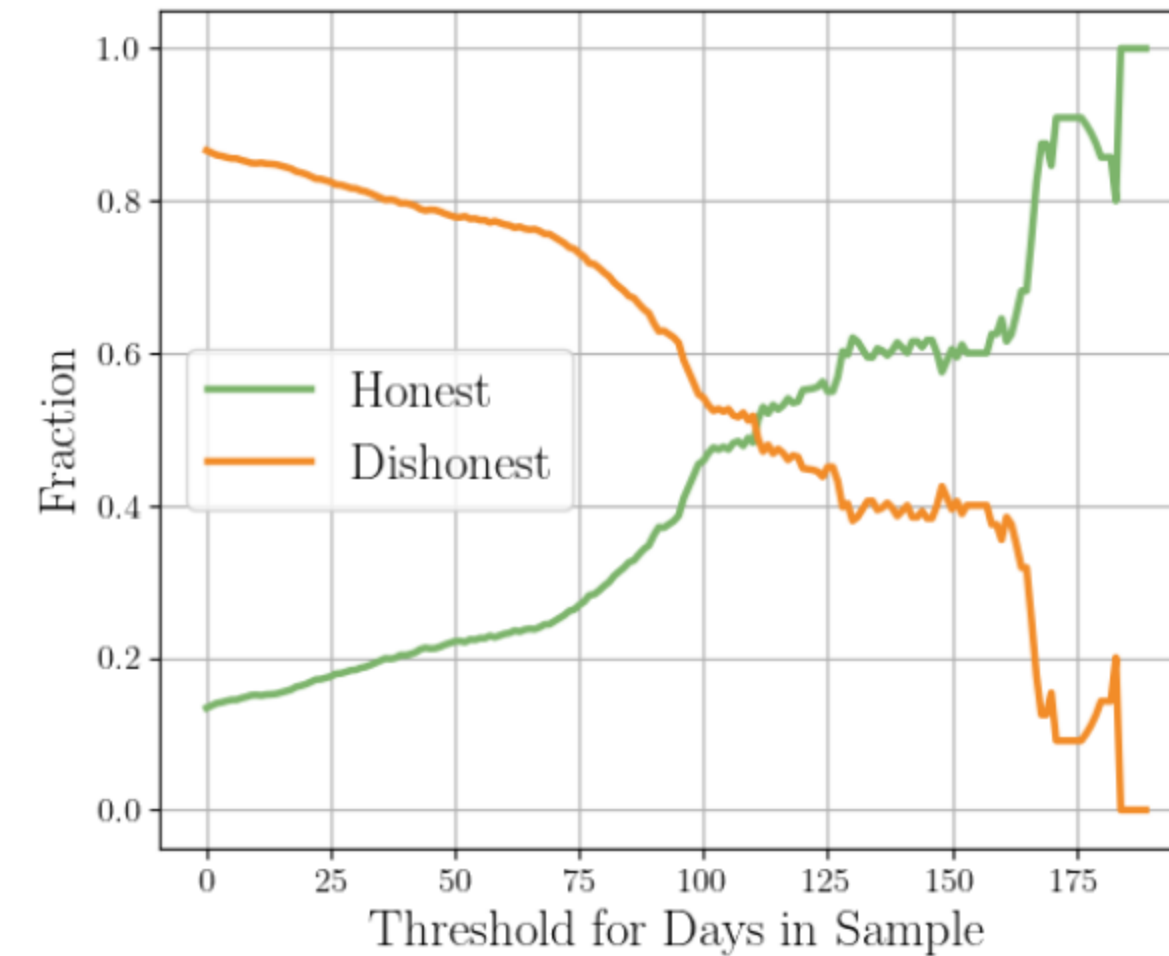
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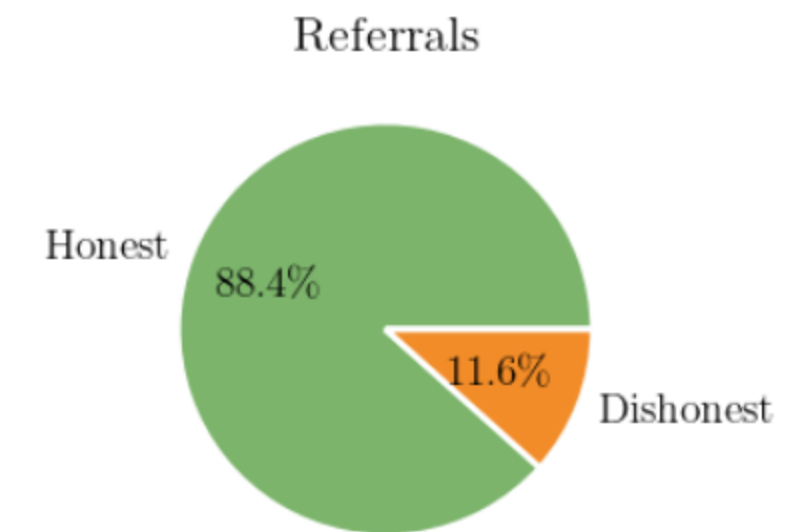
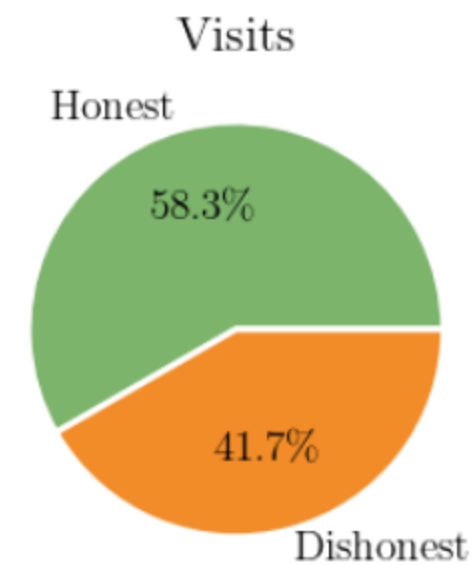
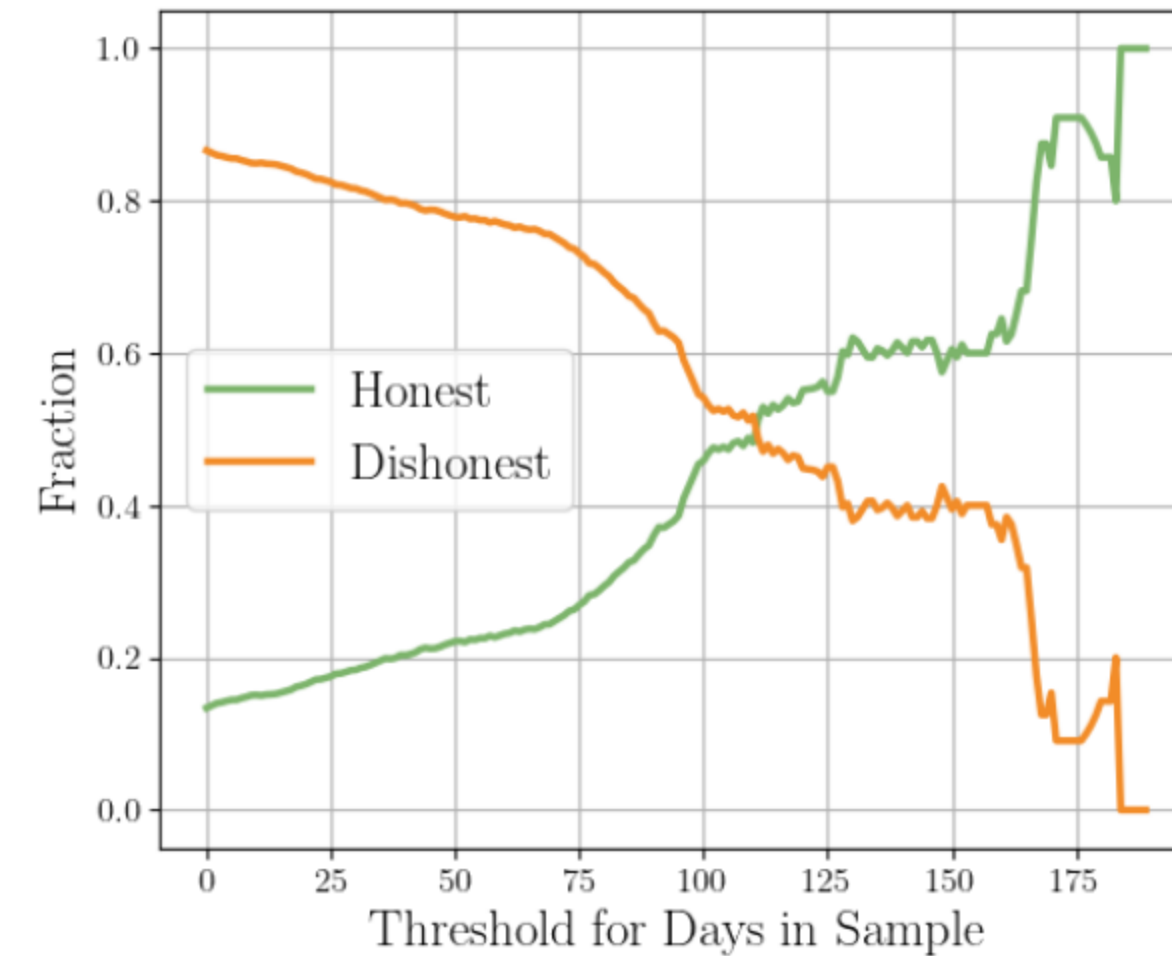
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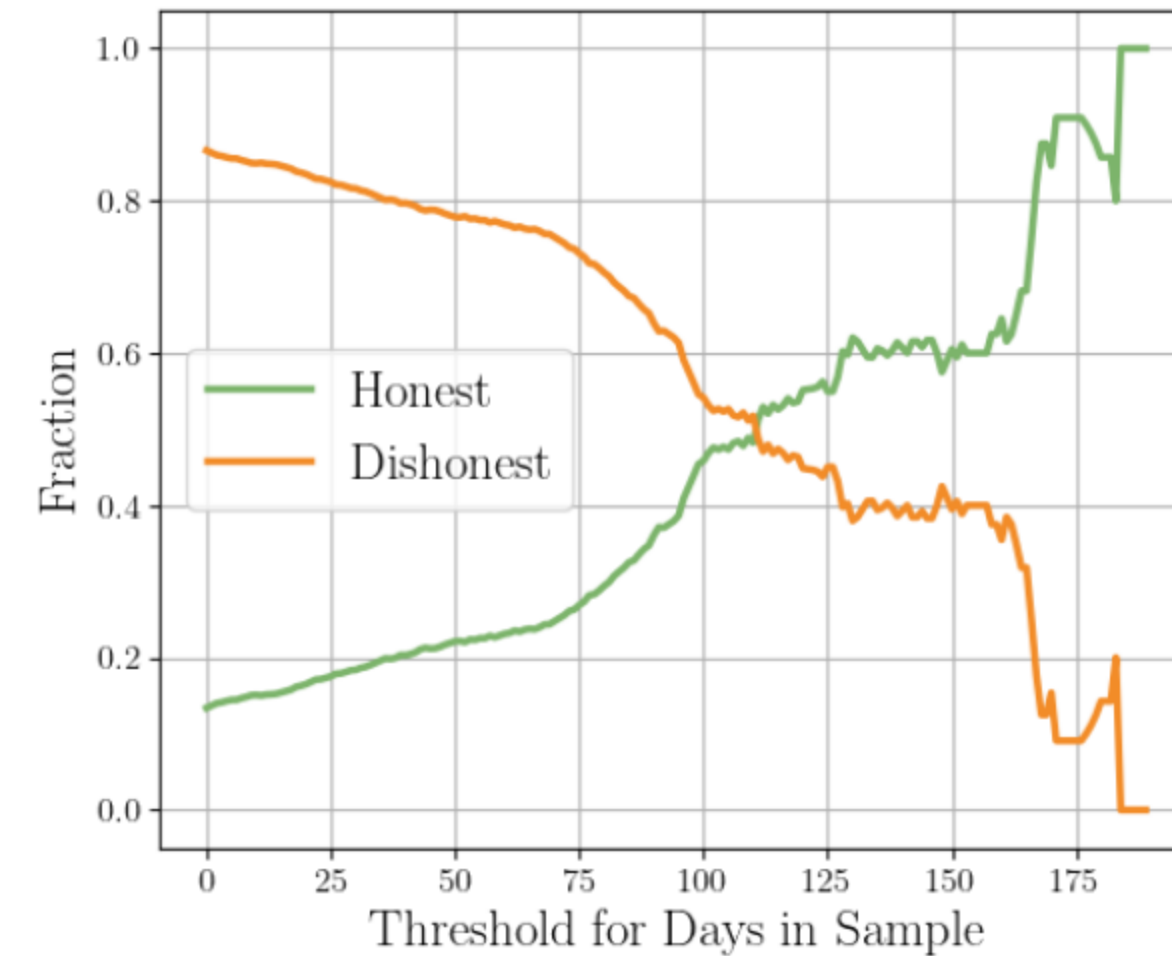
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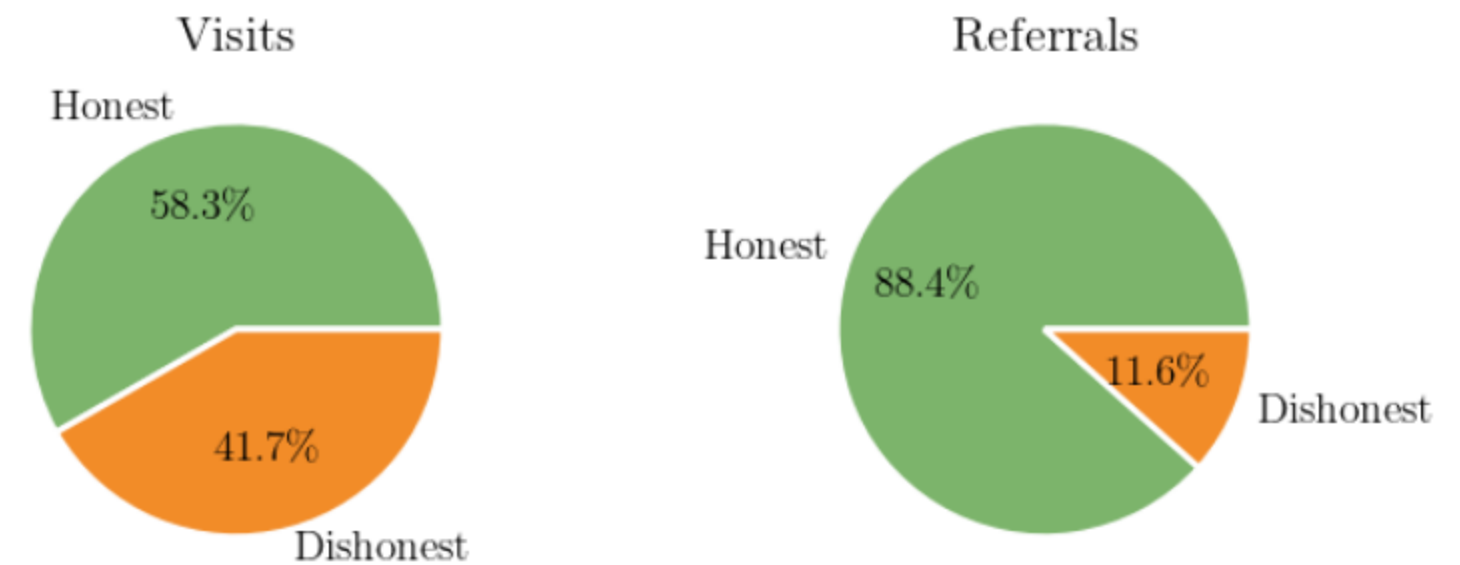


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- Validation with IP addresses and geolocation
- Results were mixed
 - Fully verified 15% of users
- Digging deeper on the data
 - Users in the sample for longer are more honest
 - Honest users contribute much richer data

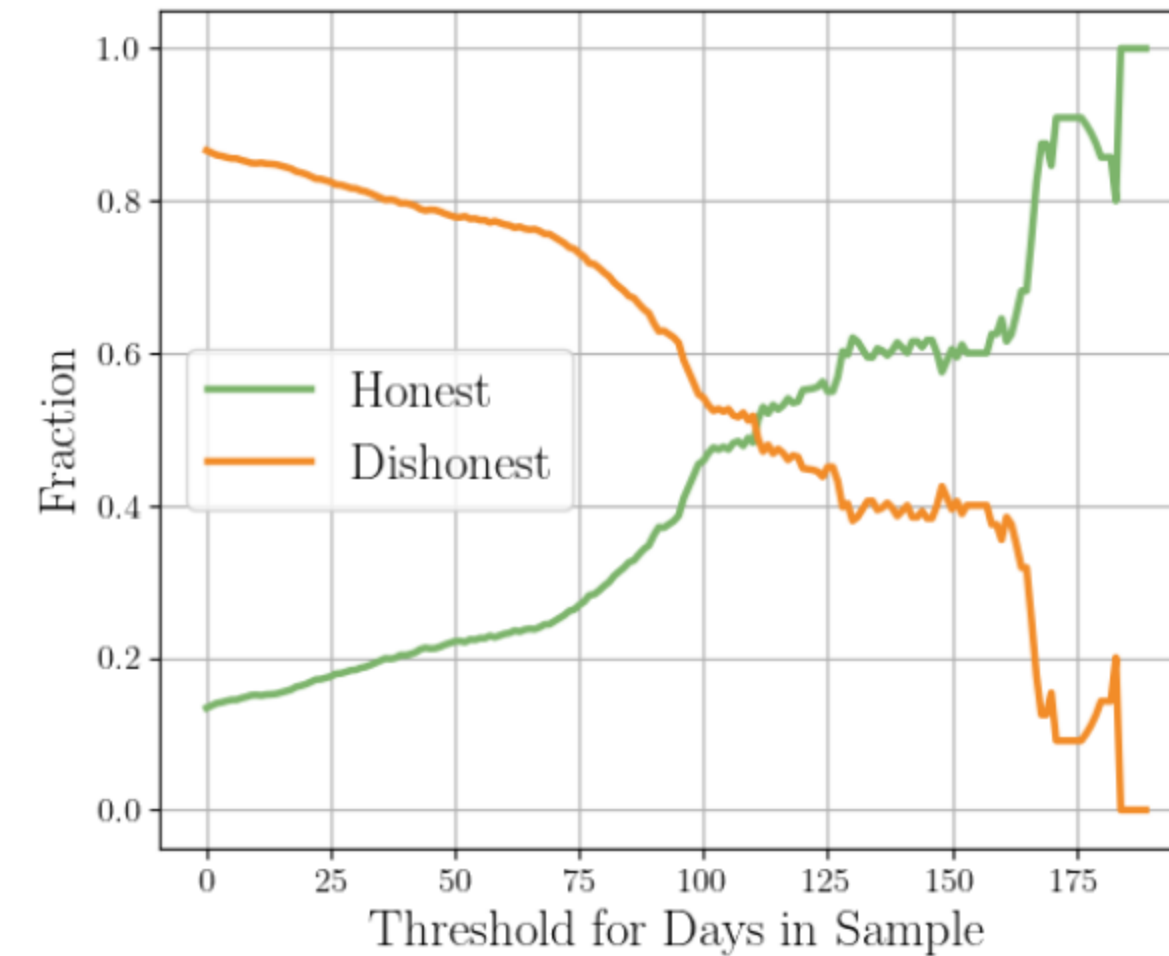


Lesson: Validating and enforcing user honesty should be a priority in future deployments.



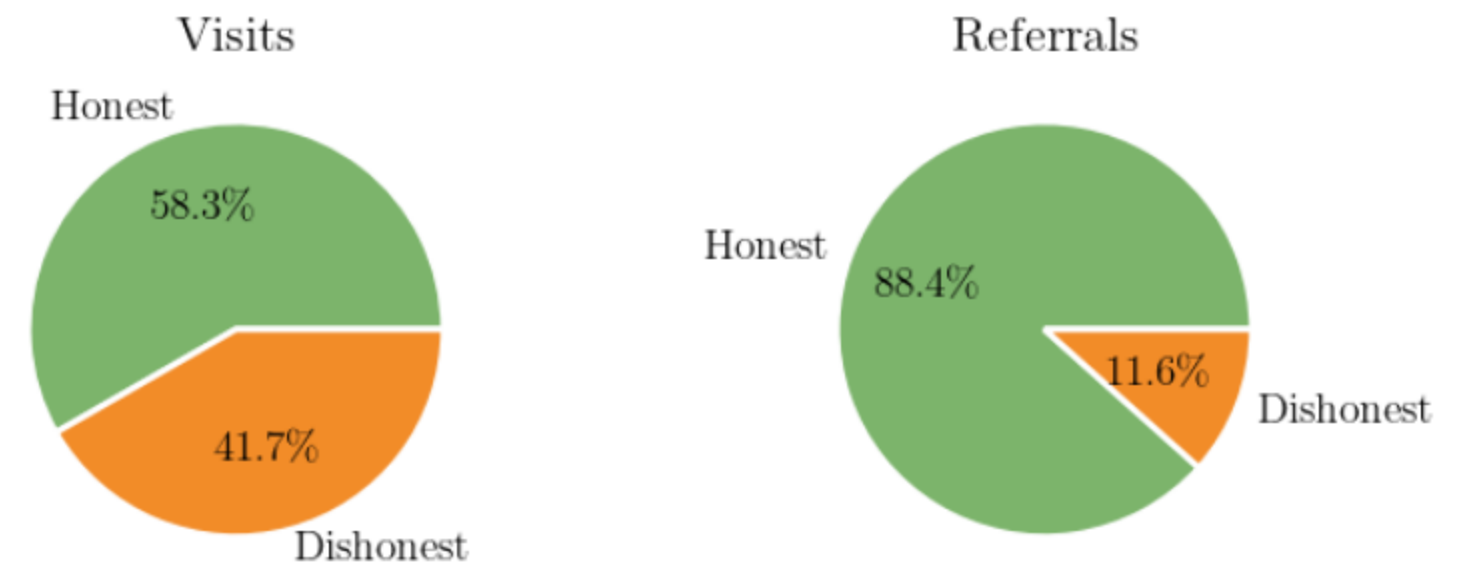
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Lesson: Our data is surprisingly robust to dishonest users.



Strengthening the Threat Model

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- Reduce or eliminate trust in the core computation

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- AWS as a single point of failure
- Reduce or eliminate trust in the core computation
- Anonymous payments

Thank You!

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Payment

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The logo for Amazon Mechanical Turk, featuring the word "amazon" in white with a curved arrow underneath it, followed by the words "mechanical turk" in orange.

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Communication by email

Data Partition

Data Partition



Data Partition



Input

Web Browsing Data

Registered IDs

Data Partition



Input

Web Browsing Data

Registered IDs

Output

Aggregation by ID

ID Set per Day

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- BU sends all collected tags to WashU
- WashU computes all combinations of users and dates and checks for matches

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 - All (**date**, **hash**) combinations
- BU aggregates data from same anonymous ID for each week